

Wireless Communication

Complete Numerical Problems & Solutions

EX 715 (BEI) · EX 751 (BEX)

Year IV Part I

Institute of Engineering, Tribhuvan University

All **46 numerical problems** from the five numerical chapters, in one volume. Every problem is a real PYQ drawn from **all 22 past papers (2070–2082 BS)** and opens with the **question exactly as the paper prints it**, so it can be attempted from this PDF alone. Solutions are worked in full — what was calculated, why, and from which formula.

Chapters 1, 4 and 8 set no numerals — no paper from 2070 to 2082 asks for one — so they do not appear here.

Theory lives in the companion volume, **Wireless Communication — Complete Exam Notes**.

Ch	Topic	Hours	Papers	Marks	Share	Numerical	Theory	Num. %
1	Introduction	2	23/23	117	5.7%	0	117	0%
2	Cellular Mobile Communication Concept	4	21/23	228	11.2%	111	117	49%
3	Radio Wave Propagation	12	23/23	447	21.9%	198	249	44%
4	Modulation and Demodulation Methods	4	21/23	209	10.3%	0	209	0%
5	Equalization and Diversity Techniques	4	22/23	212	10.4%	0	212	0%
6	Speech and Channel Coding Fundamentals	4	23/23	229	11.2%	5	224	2%
7	Multiple Access in Wireless Communications	9	23/23	312	15.3%	72	240	23%
8	Wireless Systems and Standards	6	22/23	283	13.9%	8	275	3%
Total		45		2037	100%	394	1643	19%

Share is of every mark the Detailed PYQ records. *Numerical* = a calculation, construction or trace of the kind the numerical companion solves; a question's whole marks go to one side. Bold numerical: at least half the chapter.

Contents

Ch 2 · Cellular Concepts	3
Ch 3 · Radio Wave Propagation	13
Ch 5 · Equalization and Diversity	29
Ch 6 · Speech and Channel Coding	34
Ch 7 · Multiple Access	40
Abbreviations	49

Ch 2 · Cellular Concepts Numerical Problems & Solutions

111 numerical marks, 5.4% of the archive, 49% of this chapter · Every problem below is a real PYQ, with its year tags and marks. Solutions are worked in full: what was calculated, why, and which formula was used. **Nothing is skipped to save space.**

The five formulas everything here is built on

- **Cluster size** $N = i^2 + ij + j^2$ (allowed: 1, 3, 4, 7, 9, 12, 13, 16, 19, 21)
- **Co-channel reuse ratio** $Q = D/R = \sqrt{3N}$
- **Worst-case SIR** $\frac{S}{I} = \frac{(\sqrt{3N})^n}{i_0}$, so $N = \frac{1}{3} \left(i_0 \cdot \frac{S}{I} \right)^{2/n}$
- **Hexagonal cell area** $A = \frac{3\sqrt{3}}{2} R^2 = 2.598 R^2$
- **Traffic** $A_u = \lambda H$, $A = U A_u$, and Erlang B for the GoS

i_0 = number of first-tier interferers: 6 omni, 2 for 120° sectoring, 1 for 60° sectoring. n = path loss exponent.

Problem 1 Prove $Q = \sqrt{3N}$ and find the optimal N

TOP 6/23 71 Bh **4+4**, 79 Bh **3+5**, 80 Bh **4**, 70 Bh **3+4+3**, 81 Bh **3+3**, 74 Ma **4**

1.1 SIR = 15 dB worst case, $n = 4$, considering trunking efficiency (71 Bh)

QUESTION AS PRINTED 2071 Bhadra · Q2 · [4+4]

Prove that for a hexagonal geometry the co-channel reuse ratio is given by $Q = \sqrt{3N}$; Where $N = i^2 + j^2 + ij$. A cellular service provider decides to use a digital TDMA scheme which can tolerate a Signal-to-Interference Ratio of 15 dB in the worst case. Find the optimal value of N for

- Omni directional antennas
- 120° Sectoring
- 60° Sectoring

[Use path loss exponent of 4 and consider trunking efficiency]

- **Step 1: convert SIR to a linear ratio.** The formula is in power ratio, not dB.

$$\frac{S}{I} = 10^{15/10} = 10^{1.5} = 31.6228$$

- **Step 2: apply** $N = \frac{1}{3} (i_0 \cdot S/I)^{2/n}$ for each antenna configuration. Equivalently find Q first, then $N = Q^2/3$.

Configuration	i_0	$Q = (i_0 S/I)^{1/n}$	$N = Q^2/3$	Round up to allowed N
Omni-directional	6	$(6 \times 31.6228)^{1/4} = (189.74)^{0.25} = 3.7114$	$3.7114^2/3 = 4.5915$	$N = 7$
120° sectoring	2	$(2 \times 31.6228)^{1/4} = (63.246)^{0.25} = 2.8201$	$2.8201^2/3 = 2.6509$	$N = 3$
60° sectoring	1	$(1 \times 31.6228)^{1/4} = 2.3714$	$2.3714^2/3 = 1.8745$	$N = 3$

- **Step 3: why round up, not to the nearest.** N must be one of 1, 3, 4, 7, 9, 12, ..., and a smaller N gives a worse SIR. Rounding down would break the 15 dB requirement. **So always take the next allowed value above the computed one.**
 - omni: 4.5915 → next allowed is 7 (4 is allowed but $4 < 4.5915$),
 - 120°: 2.6509 → 3,
 - 60°: 1.8745 → 3.
- **Step 4: comment on the results (the marks for “considering trunking efficiency”).**
 - Sectoring cuts the required cluster size from 7 to 3, so each cell keeps $7/3 = 2.33$ times as many channels.
 - But 60° sectoring gives the same $N = 3$ as 120° here, while splitting each cell’s channels into 6 pools instead of 3.
 - A pool of 6 smaller groups carries less traffic at the same GoS than 3 larger ones. **So 60° buys no extra capacity and loses more trunking efficiency.**
 - **Conclusion: 120° sectoring with $N = 3$ is the best choice.**

1.2 SIR = 15 dB worst case, $n = 3$ (79 Bh)

QUESTION AS PRINTED 2079 Bhadra · Q2 · [3+5]

Define Grade of Service (GoS) and explain how can it be measured in a 'blocked call cleared' type of trunking system. A cellular service provider decides to use a digital TDMA scheme that can tolerate a signal-to-interference ratio of 15 dB in the worst case. The mobile radio channel provided a propagation path loss exponent of $n = 3$. Find the optimal value of N for (a) Omni-directional antennas, (b) 1200 sectoring, and (c) 60° sectoring. Comment on your results.

"1200 sectoring" and "(c)" are the paper's own typos for 120° and (c).

- Same $S/I = 31.6228$, but now the exponent is $2/n = 2/3$.

Configuration	i_0	$Q = (i_0 S/I)^{1/3}$	$N = Q^2/3$	Round up
Omni	6	$(189.74)^{1/3} = 5.7462$	$5.7462^2/3 = 11.0064$	$N = 12$
120°	2	$(63.246)^{1/3} = 3.9842$	$3.9842^2/3 = 5.2913$	$N = 7$
60°	1	$(31.6228)^{1/3} = 3.1623$	$3.1623^2/3 = 3.3333$	$N = 4$

- **Comment.** A smaller path loss exponent means interference decays more slowly, so co-channel cells must sit further apart and N rises sharply ($7 \rightarrow 12$ for omni).
- **Sectoring is therefore even more valuable in a low- n environment:** it takes N from 12 down to 4, a $3\times$ capacity gain.

1.3 The AMPS case, SIR = 18 dB, $n = 4$ (standard textbook check)

PROBLEM not a PYQ · the standard AMPS check verified/added

An AMPS operator requires a worst-case signal-to-interference ratio of 18 dB with a path loss exponent $n = 4$. Find the smallest allowable cluster size N for omni-directional antennas.

- $S/I = 10^{1.8} = 63.0957$.
- Omni: $Q = (6 \times 63.0957)^{1/4} = (378.57)^{0.25} = 4.4110$, so $N = 4.4110^2/3 = 6.4857 \rightarrow \boxed{N = 7}$.
- **This is exactly why every early AMPS system used a 7-cell cluster.**

1.4 Exact Vs approximate SIR for $N = 7$, $n = 4$

PROBLEM not a PYQ · the follow-up every examiner likes verified/added

For a 7-cell cluster with $n = 4$, compute the worst-case S/I twice: first with the approximation that all six first-tier interferers lie at the same distance D , and then exactly, with a mobile at the cell edge seeing interferers at $D - R$, D , D , D , D and $D + R$. Compare the two, in dB.

- **Approximate** (all six interferers taken at distance D):

$$\frac{S}{I} = \frac{(\sqrt{3} \times 7)^4}{6} = \frac{4.5826^4}{6} = \frac{441.0}{6} = 73.50 = 18.66 \text{ dB}$$

- **Exact** (mobile at the cell edge, true distances $D - R$, D , D , D , D , $D + R$ with $Q = D/R = 4.5826$):

$$\frac{S}{I} = \frac{1}{2(Q-1)^{-4} + 2(Q+1)^{-4} + 2Q^{-4}} = 53.376 = 17.27 \text{ dB}$$

- **The approximation is optimistic by about 1.4 dB.** Both fall short of the 18 dB AMPS target, which is the standard justification for sectoring.

Problem 2 Spectrum, duplex channels and channels per cell

PIN 1/23 70 Bh 3+4+3

QUESTION AS PRINTED 2070 Bhadra · Q2 · [3+4+3]

Explain the difference between co-channel and adjacent channel interference. Prove that the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where $N = i^2 + ij + j^2$ is the cluster size. If 20 MHz of total spectrum is allocated for a duplex (i.e. bidirectional) wireless cellular system and each simplex (i.e. one-way) channel has 25 KHz of bandwidth, find

- The number of duplex channels, and
- The total number of channels per cell, assuming a cluster size of $N = 4$.

The $Q = \sqrt{3N}$ proof is Problem 1 above; only the spectrum arithmetic is new here.

- **Step 1: what one duplex channel actually costs.** A duplex (two-way) channel needs **two** simplex channels, one for each direction.

$$B_{\text{duplex}} = 2 \times 25 \text{ kHz} = 50 \text{ kHz}$$

Missing this factor of 2 is the single most common error in this question.

- **Step 2: total duplex channels.**

$$N_{\text{total}} = \frac{20 \times 10^6}{50 \times 10^3} = \boxed{400 \text{ duplex channels}}$$

- **Step 3: channels per cell for $N = 4$.**

$$k = \frac{S}{N} = \frac{400}{4} = \boxed{100 \text{ channels per cell}}$$

- **Step 4: the other cluster sizes, for comparison.**
 - $N = 7$: $400/7 = 57.14 \rightarrow 57$ channels per cell,
 - $N = 12$: $400/12 = 33.33 \rightarrow 33$ channels per cell.
- **Always round down here:** a fractional channel cannot be allocated.

Problem 3 Minimum co-channel distance from the cell area

PIN 1/23 74 Bh 4

QUESTION AS PRINTED 2074 Bhadra · Q2 (b) · [4]

For a seven cell reuse pattern, find the minimum distance between centers of co-channel cells. Area of each cell is uniform and is equal to 23 square km.

- **Step 1: get R from the area.** The cell is a regular hexagon of “radius” R (centre to vertex), so

$$A = \frac{3\sqrt{3}}{2} R^2 = 2.598 R^2$$

$$R = \sqrt{\frac{A}{2.598}} = \sqrt{\frac{23}{2.598}} = \sqrt{8.8530} = 2.9754 \text{ km}$$

- **Step 2: apply the reuse ratio.** For $N = 7$,

$$Q = \sqrt{3N} = \sqrt{21} = 4.5826$$

- **Step 3: minimum co-channel distance.**

$$D = Q \cdot R = 4.5826 \times 2.9754 = \boxed{13.635 \text{ km}}$$

- **Sanity check.** $D/R = 13.635/2.9754 = 4.58 \checkmark$ matches $\sqrt{21}$.
- **If the question had given the area as a circle instead,** use $A = \pi R^2$ and $R = \sqrt{23/\pi} = 2.7052 \text{ km}$, giving $D = 12.397 \text{ km}$. Read the wording carefully: “uniform cell” in this paper means the hexagon.

Problem 4 Users supported and market penetration

HOT 3/23 80 Ba, 81 Ba **5**, 79 Ch

QUESTION AS PRINTED 2080 Baishakh · Q2 (a) · [5] · 2079 Chaitra · Q2 (a) · [5]

An urban area has a population of two million residents. System A has 394 cells with 19 channels each. Find the number of users that can be supported at 2% blocking if each user average two calls per hour at average call duration of three minutes. Assuming that trunk systems are operated at maximum capacity, compute the percentage market penetration of cellular provider.

Both papers print this word for word (79 Ch reads “averages two calls”), and 79 Ch attaches the Erlang B table with the $N = 19$ row: 11.23 at 1%, 12.33 at 2%.

- **Step 1: traffic intensity per user.** $A_u = \lambda H$, and both terms must be in the *same* time unit.

$$\lambda = 2 \text{ calls/hour}, \quad H = 3 \text{ min} = \frac{3}{60} \text{ hour}$$

$$A_u = 2 \times \frac{3}{60} = \boxed{0.1 \text{ Erlang per user}}$$

- **Step 2: carried traffic per cell, from the Erlang B table.** With $C = 19$ channels and $\text{GoS} = 0.02$,

$$A = 12 \text{ Erlang}$$

(The exact Erlang B value is 12.333 Erlang; the standard table rounds to 12, which is what the marking scheme expects.)

- **Step 3: users per cell.**

$$U = \frac{A}{A_u} = \frac{12}{0.1} = 120 \text{ users per cell}$$

- **Step 4: total subscribers.**

$$U_{\text{total}} = 120 \times 394 = \boxed{47\,280 \text{ subscribers}}$$

- **Step 5: market penetration.**

$$\frac{47\,280}{2\,000\,000} = 0.02364 = \boxed{2.364 \%}$$

- **Note the assumption stated in the question:** “trunked systems operate at maximum capacity”. That is what allows the full 12 Erlang per cell to be used.
- Using the exact 12.333 Erlang instead gives 123.33 users/cell, 48 592 subscribers and 2.430 % penetration. Either is acceptable if the table value used is stated.

Problem 5 Trunking with 1 and 5 channels, then doubled load

PIN 1/23 77 Ch **2+2+2+3**

QUESTION AS PRINTED 2077 Chaitra · Q2 · [2+2+2+3]

A mobile radio system where each user averages three calls per hour and each call lasting an average of 5 minutes.

- What is traffic intensity for each user?
- Find the number of users that could use the system with 1% blocking if only one channel is available.
- Find the number of users that could use the system with 1% blocking if five trunked channels are available.
- If the number of users you found in (c) is suddenly doubled, what is the new blocking probability of the five channel trunked mobile radio system? Justify whether the performance is acceptable or not.

No. of Trunks (N)	0.1%	0.2%	0.5%	1%	1.2%	3%	5%	7%
1	0.001	0.002	0.005	0.010	0.012	0.031	0.053	0.075
2	0.046	0.065	0.105	0.153	0.168	0.282	0.381	0.470
3	0.194	0.249	0.349	0.455	0.489	0.715	0.899	1.06
4	0.439	0.535	0.701	0.869	0.922	1.26	1.52	1.75
5	0.762	0.900	1.13	1.36	1.43	1.88	2.22	2.50

Table caption as printed: “Traffic (A) in erlangs for B%”.

- **(a) Traffic intensity per user.**

$$A_u = \lambda H = 3 \times \frac{5}{60} = \boxed{0.25 \text{ Erlang}}$$

- (b) **One channel, GoS = 1 %.** From the printed table, $C = 1$ and $B = 1\%$ give $A = 0.010$ Erlang.

$$U = \frac{A}{A_u} = \frac{0.010}{0.25} = 0.04 \text{ users}$$

$$\Rightarrow \boxed{0 \text{ users}}$$

Interpretation: a single channel cannot support even one such user at 1 % blocking. A lone user offering 0.25 Erlang to one channel would be blocked about $0.25/(1 + 0.25) = 20\%$ of the time.

- (c) **Five trunked channels, GoS = 1 %.** From the table, $C = 5$ and $B = 1\%$ give $A = 1.36$ Erlang.

$$U = \frac{1.36}{0.25} = 5.44 \Rightarrow \boxed{5 \text{ users}}$$

- **This is the whole point of trunking:** 1 channel supports 0 users, but 5 channels support 5. **Capacity grows faster than the channel count.**
- (d) **Users doubled to 10.**

$$A = U A_u = 10 \times 0.25 = 2.5 \text{ Erlang}$$

Apply the Erlang B formula with $C = 5$, $A = 2.5$:

$$\text{Pr}[\text{blocking}] = \frac{A^C / C!}{\sum_{k=0}^C A^k / k!}$$

k	0	1	2	3	4	5
$A^k / k!$	1	2.5	3.125	2.6042	1.6276	0.8138

$$\sum_{k=0}^5 \frac{A^k}{k!} = 11.6706, \quad \text{Pr}[\text{blocking}] = \frac{0.8138}{11.6706} = 0.0697$$

$$\boxed{\text{GoS} = 6.97\%}$$

- **Justification.** The design target was 1 %. Doubling the users has pushed blocking to **almost 7 %, seven times worse**. **This is not acceptable:** roughly 1 call in 14 fails in the busy hour.
- **Note how non-linear this is:** doubling the load multiplied the blocking by 7, not by 2. That non-linearity is the reason cellular systems are dimensioned on the busy hour.

Problem 6 Full system design from area and spectrum

PIN 1/23 72 Ma 12 · the longest numerical in the chapter

QUESTION AS PRINTED 2072 Magh · Q2 · [12]

A city with a coverage area of 1500 sq km is covered with a 12-cell system each with a radius of 1.387 km. If the total spectrum allocated is 28.5 MHz with a full duplex channel bandwidth of 25 MHz. Assume a GOS of 0.02 for a blocked calls cleared system, is specified and the offered traffic per user is 0.03 Erlangs and traffic intensity of each cell is 84 Erlang, compute:

- the number of cells in the service area
- the number of channels per cell
- the maximum carrier traffic
- the total number of users that can be served for 2% GOS
- the number of mobiles per unique channel.
- Theoretical maximum number of users that could be served at one time by the system.

The paper prints “25 MHz” for the channel bandwidth: a typo for 25 kHz, since 28.5 MHz would otherwise buy one channel. Solve with 25 kHz and say so.

- (a) **Number of cells in the service area.**

$$A_{\text{cell}} = 2.598 R^2 = 2.598 \times (1.387)^2 = 2.598 \times 1.9238 = 4.998 \text{ km}^2$$

$$n_{\text{cells}} = \frac{1500}{4.998} = 300.12 \Rightarrow \boxed{300 \text{ cells}}$$

– (b) Channels per cell.

$$N_{\text{total}} = \frac{28.5 \times 10^6}{25 \times 10^3} = 1140 \text{ duplex channels}$$

The question says “full duplex channel bandwidth 25 kHz”, so the $\times 2$ of Problem 2 does *not* apply here. Read the wording.

$$k = \frac{1140}{12} = \boxed{95 \text{ channels per cell}}$$

- (c) Maximum carried traffic per cell. The question supplies this as **84 Erlang**. Cross-check from Erlang B: $C = 95$ at $\text{GoS} = 0.02$ gives $A = 83.13$ Erlang, so the given 84 is consistent.

$$A_{\text{cell}} = \boxed{84 \text{ Erlang}}$$

System total = $84 \times 300 = 25\,200$ Erlang.

- (d) Total users served at 2 % GoS.

$$U_{\text{cell}} = \frac{A_{\text{cell}}}{A_u} = \frac{84}{0.03} = 2800 \text{ users per cell}$$

$$U_{\text{total}} = 2800 \times 300 = \boxed{840\,000 \text{ users}}$$

- (e) Mobiles per unique channel.

$$\frac{U_{\text{cell}}}{k} = \frac{2800}{95} = \boxed{29.47 \text{ mobiles per channel}}$$

le roughly 29 subscribers share each radio channel, which is exactly the statistical multiplexing gain trunking delivers.

- (f) Theoretical maximum users served at one time. Every channel in every cell busy simultaneously:

$$95 \times 300 = \boxed{28\,500 \text{ simultaneous calls}}$$

- **Sanity check on (d) against (f):** 840 000 subscribers share 28 500 simultaneous circuits, a ratio of 29.5:1, matching (e). ✓

Problem 7 Area, channels and simultaneous calls

PIN 1/23 82 Bh 4

QUESTION AS PRINTED 2082 Bhadra · Q2 (b) · [4]

The cellular system has 32 cells, each cell has a 1.6 km radius and the system reuse factor is 7. The system is to support 336 traffic channels in total. Determine the total geographical area covered, the number of traffic channels per cell and the total number of simultaneous calls supported by this system.

- Step 1: area of one cell.

$$A_{\text{cell}} = 2.598 \times (1.6)^2 = 2.598 \times 2.56 = 6.6509 \text{ km}^2$$

- Step 2: total geographical area.

$$A_{\text{total}} = 32 \times 6.6509 = \boxed{212.83 \text{ km}^2}$$

- Step 3: channels per cell.

$$k = \frac{S}{N} = \frac{336}{7} = \boxed{48 \text{ channels per cell}}$$

- Step 4: simultaneous calls. Every cell can run all 48 of its channels at once, and there are 32 cells:

$$48 \times 32 = \boxed{1536 \text{ simultaneous calls}}$$

- **Watch the trap:** the 336 channels are the **system** total, reused by every cluster. The system supports 1536 calls, not 336, because the cluster repeats $32/7 = 4.57$ times.

Problem 8 Trunking loss from omni to 60° sectoring

PIN 1/23 81 Ba 2+2+2 · the one question where sectoring is shown to cost capacity

QUESTION AS PRINTED 2081 Baishakh · Q2 (a) · [2+2+2]

For a cluster $N = 7$ system with a blocking probability of 1% and an average call length of two minutes, find the traffic capacity loss for a blocked calls cleared system due to trunking for 57 channels when going from omnidirectional antennas to 60° sectored antennas. The average request rate per user is 1 call per hour.

GOS / Channel	0.5%	1%	2%
1	.005	.0101	.0204
9	3.333	3.783	4.345
19	10.33	11.23	12.33
54	39.47	41.51	44.00
57	42.1	44.2	46.8

– Step 1: traffic per user.

$$A_u = \lambda H = 1 \times \frac{2}{60} = 0.0333 \text{ Erlang}$$

– Step 2: omni-directional case. All 57 channels sit in **one** trunked pool. From the table, $C = 57$ at 1 % gives

$$A_{\text{omni}} = 44.2 \text{ Erlang}$$

$$U_{\text{omni}} = \frac{44.2}{0.0333} = 1326 \text{ users per cell}$$

– Step 3: 60° sectored case. The cell is split into **6 sectors**, and each sector gets its own channel group:

$$\frac{57}{6} = 9.5 \Rightarrow 9 \text{ channels per sector}$$

Round down: a channel cannot be split between sectors. From the table, $C = 9$ at 1 % gives $A = 3.783$ Erlang per sector, so

$$A_{\text{sectored}} = 6 \times 3.783 = 22.698 \text{ Erlang}$$

$$U_{\text{sectored}} = \frac{22.698}{0.0333} = 681 \text{ users per cell}$$

– Step 4: the loss.

$$\Delta A = 44.2 - 22.698 = \boxed{21.50 \text{ Erlang lost}}$$

$$\frac{21.50}{44.2} \times 100 = \boxed{48.65 \% \text{ capacity loss}}$$

– Why this happens (state this, it is where the marks are).

- Erlang B is **strongly non-linear** in the channel count. One pool of 57 channels carries 44.2 Erlang, but six pools of 9 carry only $6 \times 3.783 = 22.7$ Erlang.
- That is the **loss of trunking efficiency**: a large pool absorbs random peaks far better than several small ones.
- Three channels are also lost outright to the rounding ($57 \rightarrow 6 \times 9 = 54$).

– **The complete engineering picture**: sectoring **improves SIR**, which lets N be reduced, which gives *more* channels per cell. That gain must beat the **48.65 % trunking loss** computed here, otherwise sectoring is a net loss. This problem deliberately shows only the cost side.

Problem 9 Calls per cell per hour, and SIR

lecture-note example, same pattern as 79 Bh and 72 Ma

PROBLEM not a PYQ · lecture-note example verified/added

A cellular system has 416 radio channels available, of which 21 are control channels. The average channel holding time is 3 minutes and the blocking probability during the busy hour is 2%. The frequency reuse factor is 9. Find (i) the number of calls per cell per hour, and (ii) the signal to co-channel interference ratio in dB (take $n = 4$).

- **Step 1: voice channels.**

$$416 - 21 = 395 \text{ voice channels}$$

- **Step 2: voice channels per cell.**

$$\frac{395}{9} = 43.89 \Rightarrow 44 \text{ channels per cell (taken as 44 for the table lookup)}$$

- **Step 3: carried traffic.** From the Erlang B table, $C = 44$ at $\text{GoS} = 0.02$ gives

$$A = 34.683 \text{ Erlang}$$

- **Step 4: convert Erlang to calls.** 1 Erlang = one continuously occupied channel. With a 3-minute holding time, one Erlang carries $1/3$ call per minute.

$$\text{calls per cell per minute} = \frac{34.683}{3} = 11.561$$

$$\text{calls per cell per hour} = 11.561 \times 60 = \boxed{694 \text{ calls/cell/hour}}$$

- **Step 5: SIR for $N = 9, n = 4$.**

$$Q = \sqrt{3 \times 9} = \sqrt{27} = 5.1962$$

$$\frac{S}{I} = \frac{Q^4}{6} = \frac{729}{6} = 121.5$$

$$\frac{S}{I} [\text{dB}] = 10 \log_{10}(121.5) = \boxed{20.85 \text{ dB}}$$

- **Comment:** 20.85 dB comfortably exceeds the 18 dB AMPS requirement, so $N = 9$ is a conservative, high-quality plan. $N = 7$ would give 18.66 dB, barely adequate.

Problem 10 Cell splitting: the transmit power reduction

HOT 3/23 73 Bh, 75 Bh, 81 Bh · usually a 1 or 2 mark part inside the sectoring Vs splitting question

QUESTION AS PRINTED 2081 Bhadra · Q3 · [4] · 2075 Bhadra · Q3 · [5]

A telephone network company needs to expand its capacity based on demand on a city. A group of engineers was selected to find the solution. Among the solution sectoring and cell splitting were major technique for expansion purpose. Being a cellular planning engineer, which option do you think is best and why?

QUESTION AS PRINTED 2073 Bhadra · Q2 (a) · [5]

Explain how cell splitting and sectoring improve coverage and capacity in cellular system?

No paper prints numbers for this. The transmit-power result below is the only computable part, and it is what turns a prose answer into a full-mark one.

verified/added

- **Requirement.** The new microcell of radius $R/2$ must deliver the **same received power at its own boundary** as the old cell did at its boundary. Only then does the reuse plan carry over unchanged.
- **Step 1: write both boundary powers.**

$$P_r[\text{old boundary}] \propto P_{t1} R^{-n}, \quad P_r[\text{new boundary}] \propto P_{t2} \left(\frac{R}{2}\right)^{-n}$$

- **Step 2: set them equal.**

$$P_{t1} R^{-n} = P_{t2} \left(\frac{R}{2}\right)^{-n} \Rightarrow \frac{P_{t2}}{P_{t1}} = \left(\frac{1}{2}\right)^n$$

- **Step 3: put** $n = 4$.

$$\frac{P_{t2}}{P_{t1}} = \frac{1}{16} = 0.0625$$

$$10 \log_{10}(0.0625) = \boxed{-12.04 \text{ dB, ie reduce power by 12 dB}}$$

- **Capacity effect.** Halving R needs about **4 times as many cells** ($A \propto R^2$), so the number of clusters, and therefore the capacity, rises about **4 times** for the same spectrum.

Problem 11 Cellular Vs non-cellular channel count

PIN 1/23 82 Ba **1+3** · the cleanest statement of why the cellular idea exists at all

QUESTION AS PRINTED 2082 Baishakh · Q3 · [1+3]

Consider a single high-power transmitter that can support 40 voice channels over an area of 140 km² with the available spectrum. If this area is equally divided into seven smaller cells, each supported by lower power transmitters so that each cell supports 30% of the total voice channels, then determine

- Coverage area of each cell
- Total number of voice channels available in a cellular system compared to a non-cellular system.

- **Step 1: coverage area of each cell.** The 140 km² is divided *equally* into seven cells, so

$$A_{\text{cell}} = \frac{140}{7} = \boxed{20 \text{ km}^2}$$

- **Step 2: channels carried by one cell.** Each cell supports 30% of the **total** 40 voice channels – not 30% of a share of them:

$$C_{\text{cell}} = 0.30 \times 40 = 12 \text{ channels}$$

- **Step 3: total channels in the cellular system.**

$$C_{\text{cellular}} = 7 \times 12 = \boxed{84 \text{ voice channels}}$$

- **Step 4: compare with the non-cellular system.** The single high-power transmitter carried only its 40 channels over the whole 140 km², so

$$\frac{C_{\text{cellular}}}{C_{\text{non-cellular}}} = \frac{84}{40} = \boxed{2.1 \text{ times, ie 44 extra channels}}$$

- **This is frequency reuse in one line:** the same 40 channels, re-used across seven lower-power cells, carry more than twice the traffic **without one extra hertz of spectrum**. The cost is seven base stations instead of one, plus handoff.
- **Watch the 30 %.** Taking it as 30 % of 40/7 instead of 30 % of 40 gives $7 \times 1.71 = 12$ channels total, which would make the cellular system *worse* than the single transmitter – a clear sign the reading is wrong.

Erlang B reference table

Values recomputed from $\text{Pr}[\text{blocking}] = \frac{A^C / C!}{\sum_{k=0}^C A^k / k!}$ and cross-checked against the tables printed on the 77 Ch and 81 Ba papers. **Both printed tables agree with this one** once their differing row labels (“Trunks N ” Vs “GOS / Channel”) are read correctly.

Channels C	A at 0.5 %	A at 1 %	A at 2 %	A at 5 %
1	0.005	0.010	0.020	0.053
2	0.105	0.153	0.223	0.381
3	0.349	0.455	0.602	0.899
4	0.701	0.869	1.092	1.525
5	1.132	1.361	1.657	2.219
9	3.333	3.783	4.345	5.370
10	3.961	4.461	5.084	6.216
19	10.331	11.230	12.333	14.315
20	11.092	12.031	13.182	15.249
44	30.797	32.543	34.682	38.557
54	39.474	41.505	43.997	48.536
57	42.109	44.222	46.816	51.548
95	76.325	79.368	83.133	90.123

How to use it in the exam: you are always given either C and the GoS (look up A , then divide by A_u for users), or C and A (look across the row for the GoS). Never try to sum the series by hand unless $C \leq 5$.

Ch 3 · Radio Wave Propagation Numerical Problems & Solutions

198 numerical marks, 9.7% of the archive, 44% of this chapter · This is the heaviest numerical chapter in the subject. A propagation numerical has appeared in every one of the 23 papers. Every problem below is a real PYQ, worked in full.

The formula sheet

- **Free space (Friis)** $P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$, $\lambda = c/f$
- **Free space path loss** $PL = 20 \log \left(\frac{4\pi d}{\lambda} \right) = 32.44 + 20 \log f_{\text{MHz}} + 20 \log d_{\text{km}}$
- **Power flux density** $P_d = \frac{P_t G_t}{4\pi d^2} = \frac{|E|^2}{120\pi}$, **aperture** $A_e = \frac{G_r \lambda^2}{4\pi}$, $P_r = P_d A_e$
- **Matched receiver** $V_{\text{rms}} = \sqrt{P_r R_{\text{ant}}}$, open-circuit $V_{\text{ant}} = 2V_{\text{rms}}$
- **Two-ray** $P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4}$, $PL = 40 \log d - 20 \log h_t - 20 \log h_r$ (unity gains)
- **Log-distance** $PL(d) = PL(d_0) + 10n \log \left(\frac{d}{d_0} \right) + X_\sigma$
- **Okumura** $L_{50} = L_F + A_{mu} - G(h_{te}) - G(h_{re}) - G_{\text{AREA}}$, with $G(h_{te}) = 20 \log \frac{h_{te}}{200}$, $G(h_{re}) = 10 \log \frac{h_{re}}{3}$ for $h_{re} \leq 3$ m, $20 \log \frac{h_{re}}{3}$ for $3 < h_{re} < 10$ m
- **Hata (urban)** $L_{50} = 69.55 + 26.16 \log f_c - 13.82 \log h_{te} - a(h_{re}) + (44.9 - 6.55 \log h_{te}) \log d$
- **Delay spread** $\bar{\tau} = \frac{\sum P_k \tau_k}{\sum P_k}$, $\bar{\tau}^2 = \frac{\sum P_k \tau_k^2}{\sum P_k}$, $\sigma_\tau = \sqrt{\bar{\tau}^2 - \bar{\tau}^2}$
- **Coherence** $B_c \approx \frac{1}{50\sigma_\tau}$ (90%), $B_c \approx \frac{1}{5\sigma_\tau}$ (50%), $f_m = \frac{v}{\lambda}$, $T_c \approx \frac{0.423}{f_m}$

Units discipline, the source of most lost marks: P in W for Friis, dBm only after conversion. d in metres in $20 \log(4\pi d/\lambda)$ but in km in the 32.44 form and in Hata. f in MHz in the 32.44 form and in Hata. Always write the unit beside every number.

Problem 1 Free space: received power, E-field, flux density, rms voltage

HOT 3/23 80 Ba, 81 Ba **8**, 78 Ch **2+2+2+2** · the classic four-part free space numerical

QUESTION AS PRINTED 2080 Baishakh · Q3 (b) · [8]

Assume free space propagation, a receiver is located 10 km away from a 50 w transmitter. The carrier frequency is 900 MHz. antenna gain at transmitter and receiver are 1 and 2 respectively, find:

- The power received at the receiver
- The magnitude of E-field at the receiver antenna
- The power flux density
- The rms voltage applied to the receiver input.

The receiver antenna has a purely real impedance of 50 Ω and is matched to the receiver.

QUESTION AS PRINTED 2078 Chaitra · Q3 (b) · [2+2+2+2]

Assume free space propagation, a receiver is located 10 km away from a 50W transmitter. The carrier frequency is 900 MHz, antenna gain at transmitter and receiver is 1 and 2 respectively. Find: (i) The power received at the receiver. (ii) The magnitude of the E-field at the receiver. (iii) The rms voltage applied to the receiver input. (iv) The power flux density.

Same numbers, parts (iii) and (iv) swapped. "900 MHz" is the 80 Ba paper's typo.

Step 0: wavelength

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{900 \times 10^6} = 0.3333 \text{ m}$$

(i) Received power

- Friis, with $L = 1$ (no system loss stated):

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2} = \frac{50 \times 1 \times 2 \times (0.3333)^2}{(4\pi)^2 (10^4)^2}$$

$$= \frac{50 \times 2 \times 0.11111}{157.91 \times 10^8} = \frac{11.111}{1.5791 \times 10^{10}} = 7.0362 \times 10^{-10} \text{ W}$$

– In dBm:

$$P_r = 10 \log_{10}(7.0362 \times 10^{-10} \times 10^3) = \boxed{-61.53 \text{ dBm}} = -91.53 \text{ dBW}$$

(iii) Power flux density (do this before the E-field, it feeds it)

$$P_d = \frac{P_t G_t}{4\pi d^2} = \frac{50 \times 1}{4\pi (10^4)^2} = \frac{50}{1.2566 \times 10^9} = \boxed{3.9789 \times 10^{-8} \text{ W/m}^2}$$

(ii) Magnitude of the E-field

– Free space intrinsic impedance $\eta = 120\pi = 377 \Omega$, and $P_d = |E|^2/\eta$:

$$\begin{aligned} |E| &= \sqrt{P_d \times 377} = \sqrt{3.9789 \times 10^{-8} \times 377} = \sqrt{1.5 \times 10^{-5}} \\ &= \boxed{3.873 \times 10^{-3} \text{ V/m}} = 3.873 \text{ mV/m} \end{aligned}$$

Cross-check: P_r from P_d and the aperture

$$\begin{aligned} A_e &= \frac{G_r \lambda^2}{4\pi} = \frac{2 \times 0.11111}{12.566} = 0.017684 \text{ m}^2 \\ P_r &= P_d A_e = 3.9789 \times 10^{-8} \times 0.017684 = 7.036 \times 10^{-10} \text{ W} \quad \checkmark \text{ matches (i)} \end{aligned}$$

(iv) rms voltage at the receiver input

– The antenna is **matched** to a purely real 50Ω load, so the power delivered is

$$\begin{aligned} P_r &= \frac{V_{\text{rms}}^2}{R_{\text{ant}}} \Rightarrow V_{\text{rms}} = \sqrt{P_r R_{\text{ant}}} \\ &= \sqrt{7.0362 \times 10^{-10} \times 50} = \sqrt{3.5181 \times 10^{-8}} \\ &= \boxed{1.8757 \times 10^{-4} \text{ V}} = 187.57 \mu\text{V} \end{aligned}$$

– If the question asks for the open-circuit antenna voltage V_{ant} instead, the matched load halves it, so

$$V_{\text{ant}} = 2V_{\text{rms}} = 375.13 \mu\text{V}$$

Read the wording: “applied to the receiver input” is V_{rms} ; “induced in the antenna” is V_{ant} .

Problem 2 Free space link with dBi antennas

PIN 1/23 80 Bh 6

QUESTION AS PRINTED 2080 Bhadra · Q2 (a) · [6]

A wireless communication transmitter has an output power of 165 watts at a carrier frequency of 325 MHz. It is connected to an antenna with a gain of 12 dBi. The receiving antenna is 15 km away and has a gain of 6 dBi. Calculate the power delivered to the receiver, considering free-space propagation. (Assume that there are no other losses or mismatches in the system.)

- **Gains are in dB, so work the whole problem in dB.** Converting them to linear and using Friis directly also works but is slower and more error-prone.
- **Step 1: transmit power in dBm.**

$$P_t = 10 \log_{10}(165 \times 10^3) = 10 \log_{10}(1.65 \times 10^5) = 52.17 \text{ dBm}$$

- **Step 2: free space path loss.** Using the km / MHz form:

$$\begin{aligned} PL &= 32.44 + 20 \log_{10}(325) + 20 \log_{10}(15) \\ &= 32.44 + 50.24 + 23.52 = 106.20 \text{ dB} \end{aligned}$$

Check with $\lambda = 3 \times 10^8 / 325 \times 10^6 = 0.9231 \text{ m}$: $20 \log(4\pi \times 15000 / 0.9231) = 106.20 \text{ dB} \checkmark$

– Step 3: link equation.

$$P_r = P_t + G_t + G_r - PL = 52.17 + 12 + 6 - 106.20$$

$$= \boxed{-36.03 \text{ dBm}}$$

– Step 4: convert back to watts if asked.

$$P_r = 10^{-36.03/10} \text{ mW} = 2.4966 \times 10^{-4} \text{ mW} = \boxed{249.7 \text{ nW}}$$

Problem 3 Full link budget with feeder and mismatch loss

PIN 1/23 72 Ash 8

QUESTION AS PRINTED 2072 Ashwin · Q3 · [8]

A BS transmitter has a power output of 10 watts operating at a frequency of 250 MHz. The transmitter is connected by 20 m of an RF coaxial cable, which has a loss of 3-dB/100 m specification, to an antenna that has a gain of 9 dBi. The receiving antenna is 25 km away and has a gain of 4 dBi. There is negligible loss in the receiver feeder line, but the receiver is mismatched; the receiving antenna and feeder cable are designed for 50 ohm impedance. The receiver impedance loss due to mismatch is of about 0.2 dB. Calculate the power delivered to the receiver, assuming free-space propagation.

– Step 1: transmit power in dBm.

$$P_t = 10 \log_{10}(10 \times 10^3) = 40.00 \text{ dBm}$$

– Step 2: transmit feeder loss.

$$L_{\text{coax}} = 20 \text{ m} \times \frac{3 \text{ dB}}{100 \text{ m}} = 0.60 \text{ dB}$$

– Step 3: free space path loss at 25 km, 250 MHz.

$$PL = 32.44 + 20 \log_{10}(250) + 20 \log_{10}(25)$$

$$= 32.44 + 47.96 + 27.96 = 108.36 \text{ dB}$$

– Step 4: assemble the budget. Every gain adds, every loss subtracts.

Item	Value	Note
Transmitter output	+40.00 dBm	10 W
Transmit feeder (coax)	−0.60 dB	20 m at 3 dB/100 m
Transmit antenna gain	+9.00 dB	9 dBi
Free space path loss	−108.36 dB	25 km, 250 MHz
Receive antenna gain	+4.00 dB	4 dBi
Receive feeder loss	−0.00 dB	stated negligible
Receiver mismatch loss	−0.20 dB	50 Ω system
Power at receiver	−56.16 dBm	

$$P_r = 40.00 - 0.60 + 9.00 - 108.36 + 4.00 - 0.20 = \boxed{-56.16 \text{ dBm}}$$

$$= 10^{-56.16/10} \text{ mW} = 2.42 \times 10^{-6} \text{ mW} = \boxed{2.42 \text{ nW}}$$

Marking tip: draw the budget as a table like this one. Examiners give method marks for a laid-out budget even when one entry is wrong.

Problem 4 Radio coverage range from a receiver threshold

PIN 1/23 70 Bh 6

QUESTION AS PRINTED 2070 Bhadra · Q4 · [6]

Determine the radio coverage range of a base station that transmits a RF signal at 150 W, given the receiver threshold level is -104 dBm. Assume that the path loss at the first meter is 15 dB in a mobile radio propagation condition. (Path loss exponent = 4)

- **Step 1: transmit power in dBm.**

$$P_t = 10 \log_{10}(150 \times 10^3) = 51.76 \text{ dBm}$$

- **Step 2: maximum allowable path loss.** The link works as long as $P_r \geq -104$ dBm, so

$$PL_{\max} = P_t - P_{r,\min} = 51.76 - (-104) = 155.76 \text{ dB}$$

- **Step 3: log-distance model with $d_0 = 1$ m.**

$$PL(d) = PL(d_0) + 10n \log_{10}\left(\frac{d}{d_0}\right) = 15 + 40 \log_{10}(d)$$

- **Step 4: solve for d .**

$$\begin{aligned} 155.76 &= 15 + 40 \log_{10}(d) \\ \log_{10}(d) &= \frac{155.76 - 15}{40} = \frac{140.76}{40} = 3.519 \\ d &= 10^{3.519} = \boxed{3304 \text{ m} \approx 3.30 \text{ km}} \end{aligned}$$

- **Sanity check:** at 3.3 km with $n = 4$ the loss beyond 1 m is $40 \log(3304) = 140.8$ dB, plus the 15 dB first-metre loss gives 155.8 dB, which is exactly the budget. ✓

Problem 5 Monopole aperture and two-ray received power

PIN 1/23 71 Bh 6

QUESTION AS PRINTED 2071 Bhadra · Q4 · [6]

A mobile is located 5 km away from a base station and a vertical $\lambda/4$ monopole antenna with a gain of 2.55 dB to receive cellular radio signals. The electric field at 1 km from the transmitter is measured to be 10^{-3} V/m. The carrier frequency used for this system is 900 MHz.

- Find the length and effective aperture of the receiving antenna.
- Find the received power at the mobile using two ray ground reflection model assuming the height of the transmitting antenna is 50 m and the receiving antenna is 1.5 m above ground.

(a) Length and effective aperture

–

$$\lambda = \frac{3 \times 10^8}{900 \times 10^6} = 0.3333 \text{ m}$$

- **Length of a quarter-wave monopole:**

$$L = \frac{\lambda}{4} = \frac{0.3333}{4} = \boxed{0.0833 \text{ m} = 8.33 \text{ cm}}$$

- **Gain in linear terms:**

$$G_r = 10^{2.55/10} = 1.7989$$

- **Effective aperture:**

$$A_e = \frac{G_r \lambda^2}{4\pi} = \frac{1.7989 \times 0.11111}{12.566} = \boxed{0.0159 \text{ m}^2}$$

(b) Received power from the two-ray model

- **Step 1: the two-ray E-field.** For d well beyond the crossover distance,

$$|E_{\text{tot}}(d)| = \frac{2E_0 d_0}{d} \cdot \frac{2\pi h_t h_r}{\lambda d}$$

with the reference $E_0 d_0 = 10^{-3} \times 1000 = 1 \text{ V}$ (measured at $d_0 = 1 \text{ km}$).

– **Step 2: substitute.**

$$\begin{aligned}
 |E_{\text{tot}}| &= \frac{2 \times 1}{5000} \cdot \frac{2\pi \times 50 \times 1.5}{0.3333 \times 5000} \\
 &= 4 \times 10^{-4} \times \frac{471.24}{1666.7} = 4 \times 10^{-4} \times 0.28274 \\
 &= \boxed{1.131 \times 10^{-4} \text{ V/m}}
 \end{aligned}$$

– **Step 3: convert field to power through the aperture.**

$$\begin{aligned}
 P_r &= \frac{|E_{\text{tot}}|^2}{120\pi} \cdot A_e = \frac{(1.131 \times 10^{-4})^2}{377} \times 0.0159 \\
 &= \frac{1.2791 \times 10^{-8}}{377} \times 0.0159 = 3.3928 \times 10^{-11} \times 0.0159 \\
 &= \boxed{5.397 \times 10^{-13} \text{ W} = -92.68 \text{ dBm}}
 \end{aligned}$$

- **Check the model is valid here:** crossover distance $d_c = 4\pi h_t h_r / \lambda = 4\pi(50)(1.5)/0.3333 = 2827 \text{ m}$. Since $d = 5000 \text{ m} > d_c$, the $1/d^2$ field approximation holds. ✓

Problem 6 Free space path loss of a cordless phone

HOT 2/23 77 Ch 6, 80 Ch 2+4

QUESTION AS PRINTED 2077 Chaitra · Q3 · [6]

Given a cordless phone operating at 52000 kHz frequency which has a range of 50m. Determine the free space path loss.

QUESTION AS PRINTED 2080 Chaitra · Q4 (b) · [2+4]

Define Doppler's effect in Small Scale Fading. (i) Given a cordless phone operating at 52000 kHz frequency which has a range of 50m. Determine the free space path loss.

– **Step 1: sort out the units.**

$$f = 52000 \text{ kHz} = 52 \text{ MHz} = 52 \times 10^6 \text{ Hz}$$

The “52000 kHz” is deliberately awkward. Convert it first.

– **Step 2: wavelength.**

$$\lambda = \frac{3 \times 10^8}{52 \times 10^6} = 5.769 \text{ m}$$

– **Step 3: path loss.**

$$\begin{aligned}
 PL &= 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right) = 20 \log_{10} \left(\frac{4\pi \times 50}{5.769} \right) \\
 &= 20 \log_{10}(108.9) = \boxed{40.74 \text{ dB}}
 \end{aligned}$$

– **Cross-check with the km / MHz form, $d = 0.05 \text{ km}$:**

$$PL = 32.44 + 20 \log(52) + 20 \log(0.05) = 32.44 + 34.32 - 26.02 = 40.74 \text{ dB} \quad \checkmark$$

- **Note how low this is** compared with the 100 dB-plus figures elsewhere: short range and a long wavelength both help. That is why cordless phones need only milliwatts.

Problem 7 dBm, dBW and received power at two distances

PIN 1/23 81 Ba 1+1+2+2

QUESTION AS PRINTED 2081 Baishakh · Q3 (b) · [1+1+2+2]

If a transmitter produces 50 watts of power, express the transmit power in units of (i) dBm, and (ii) dBW. If 50 watts is applied to a unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna. What is P_r (10 km)? Assume unity gain for the receiver antenna.

– (i)

$$P_t [\text{dBm}] = 10 \log_{10}(50 \times 10^3) = 10 \log_{10}(5 \times 10^4) = \boxed{46.99 \text{ dBm}}$$

– (ii)

$$P_t [\text{dBW}] = 10 \log_{10}(50) = \boxed{16.99 \text{ dBW}}$$

Always dBm = dBW + 30. ✓

– (iii) At $d = 100 \text{ m}$, $\lambda = 0.3333 \text{ m}$, $G_t = G_r = 1$:

$$P_r = \frac{50 \times 1 \times 1 \times (0.3333)^2}{(4\pi)^2 (100)^2} = \frac{5.5556}{157.91 \times 10^4} = 3.5181 \times 10^{-6} \text{ W}$$

$$= 10 \log_{10}(3.5181 \times 10^{-3} \text{ mW}) = \boxed{-24.54 \text{ dBm}}$$

– (iv) At $d = 10 \text{ km}$, use the $1/d^2$ scaling instead of recomputing:

$$P_r(10 \text{ km}) = P_r(100 \text{ m}) \left(\frac{100}{10000} \right)^2 = 3.5181 \times 10^{-6} \times 10^{-4}$$

$$= 3.5181 \times 10^{-10} \text{ W} = \boxed{-64.54 \text{ dBm}}$$

– **The shortcut in dB:** the distance grew by a factor 100, ie 2 decades, and free space costs **20 dB per decade**, so $-24.54 - 40 = -64.54 \text{ dBm}$. ✓

Problem 8 Far-field (Fraunhofer) distance

PIN 1/23 79 Ch 3+3

QUESTION AS PRINTED 2079 Chaitra · Q4 (b) · [3+3]

Explain basic propagation models. Compute the far field distance for an antenna with maximum dimension of 1 m and operating frequency of 900 MHz.

–

$$\lambda = \frac{3 \times 10^8}{900 \times 10^6} = 0.3333 \text{ m}$$

–

$$d_f = \frac{2D^2}{\lambda} = \frac{2 \times (1)^2}{0.3333} = \boxed{6 \text{ m}}$$

– **Verify both side conditions:**

– $d_f \gg D$: $6 \gg 1$ ✓

– $d_f \gg \lambda$: $6 \gg 0.3333$ ✓

– **Meaning:** the Friis equation may only be used beyond 6 m from this antenna. Closer in, the wavefront is not yet planar and the formula is meaningless.

Problem 9 Okumura model: median path loss and received power

HOT 5/23 74 Ma **4+4**, 80 Bh **4+2+2**, 82 Bh **5**, 82 Ba **8** · the single most repeated numerical in the chapter

QUESTION AS PRINTED 2080 Bhadra · Q3 (b) · [4+2+2]

Determine the mean path loss using Okumura's model for $d = 50$ km, $h_{te} = 100$ m, $h_{re} = 10$ m in a suburban environment. If the base station transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz. Find EIRP (dBm) and the power at the receiver where the gain at receiving antenna is 10 dB. The medium attenuation factor is 43 dB and the gain due to area is 9 dB for a 50 km distance and 900 MHz frequency of operation.

QUESTION AS PRINTED 2082 Bhadra · Q3 (b) · [5]

Find the mean path loss using Okumura's Model for $d = 50$ km, $h_{te} = 100$ m, $h_{re} = 10$ m in a suburban environment. If the BS transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz, find EIRP in dBm and the power at the receiver where gain of receiving antenna is 10 dB. (Given: $A_{mu}(f, d) = 43$ dB, $G_{Area}(f) = 9$ dB)

QUESTION AS PRINTED 2074 Magh · Q3 · [4+4]

Explain Okumura model of outdoor radio propagation. Determine the median path loss for T-R separation of 50 km, transmit antenna effective height of 100 m and receive antenna effective height of 10 m in a suburban area with correction factor of 9 dB at 900 MHz. Assume median attenuation relative to free space 43 dB.

QUESTION AS PRINTED 2082 Baishakh · Q4 · [8]

Find the median path loss using Okumura's model for $d = 50$ km, $h_{te} = 100$ m, $h_{re} = 10$ m in a suburban environment. If the base station transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz, find the power at the receiver (assume a unity gain receiving antenna). [Assume $A_{mu} = 43$ dB, $G_{AREA} = 9$ dB]

One set of numbers, four papers. "EIRP of 1 kW" is the 82 Bh paper's own typo, and "path loss sing" is 82 Ba's. **82 Ba is the only one with a unity-gain receiver**, and it does not ask for EIRP in dBm — see Step 4.

Step 1: free space path loss at 50 km, 900 MHz

$$\lambda = 0.3333 \text{ m}$$

$$L_F = 20 \log_{10} \left(\frac{4\pi \times 50 \times 10^3}{0.3333} \right) = 20 \log_{10}(1.885 \times 10^6) = 125.51 \text{ dB}$$

Check: $32.44 + 20 \log(900) + 20 \log(50) = 32.44 + 59.08 + 33.98 = 125.50 \text{ dB} \checkmark$

Step 2: antenna height gain factors

- Base station, $h_{te} = 100$ m (valid, since $30 < 100 < 1000$ m):

$$G(h_{te}) = 20 \log_{10} \left(\frac{100}{200} \right) = 20 \log_{10}(0.5) = -6.02 \text{ dB}$$

Negative, because 100 m is below the 200 m reference.

- Mobile, $h_{re} = 10$ m. Since $3 < h_{re} < 10$ m the **20 log** branch applies:

$$G(h_{re}) = 20 \log_{10} \left(\frac{10}{3} \right) = 20 \log_{10}(3.333) = +10.46 \text{ dB}$$

Choosing the wrong branch here is the standard error. Check h_{re} against 3 m first.

Step 3: assemble the Okumura result

$$\begin{aligned} L_{50} &= L_F + A_{mu} - G(h_{te}) - G(h_{re}) - G_{AREA} \\ &= 125.51 + 43 - (-6.02) - (10.46) - 9 \\ &= 125.51 + 43 + 6.02 - 10.46 - 9 = \boxed{155.07 \text{ dB}} \end{aligned}$$

Step 4 (80 Bh, 82 Bh): EIRP and received power

$$\text{EIRP} = 1 \text{ kW} = 1000 \text{ W} \Rightarrow 10 \log_{10}(1000 \times 10^3) = \boxed{60 \text{ dBm}}$$

$$P_r = \text{EIRP} - L_{50} + G_r = 60 - 155.07 + 10 = \boxed{-85.07 \text{ dBm}}$$

-

$$= 10^{-85.07/10} \text{ mW} = 3.11 \times 10^{-9} \text{ mW} = 3.11 \text{ pW}$$

- **Interpretation:** -85 dBm is a usable but not comfortable level. Typical GSM sensitivity is around -104 dBm , so this link has about 19 dB of margin.
- **82 Ba variant – unity gain receiver.** That paper asks only for P_r , with $G_r = 1$, ie **0 dB**:

$$P_r = 60 - 155.07 + 0 = \boxed{-95.07 \text{ dBm}} = 0.311 \text{ pW}$$

Exactly 10 dB below the 80 Bh / 82 Bh answer, since the only change is dropping the 10 dB receive gain. Margin over -104 dBm falls to about 9 dB .

Problem 10 Okumura run backwards: find the T-R separation

PIN 1/23 77 Ch 8

QUESTION AS PRINTED 2077 Chaitra · Q4 (b) · [8]

Employing the Okumura model compute the transmitter and receiver separation distance if median loss is 167 dB when the carrier frequency is 2.1 GHz . Assume height of transmitting antenna is 40 m , height of receiving antenna is 2 m , for a large city. [$A_{mu} = 34 \text{ dB}$, $G_{area} = 0 \text{ dB}$]

- **Step 1: the height gain factors.**

$$G(h_{te}) = 20 \log_{10} \left(\frac{40}{200} \right) = 20 \log_{10}(0.2) = -13.98 \text{ dB}$$

$$h_{re} = 2 \text{ m} \leq 3 \text{ m} \Rightarrow G(h_{re}) = 10 \log_{10} \left(\frac{2}{3} \right) = -1.76 \text{ dB}$$

Here the 10 log branch applies. Compare with Problem 9, where it was 20 log.

- **Step 2: rearrange Okumura for L_F .**

$$L_{50} = L_F + A_{mu} - G(h_{te}) - G(h_{re}) - G_{area}$$

$$L_F = L_{50} - A_{mu} + G(h_{te}) + G(h_{re}) + G_{area}$$

$$= 167 - 34 + (-13.98) + (-1.76) + 0 = \boxed{117.26 \text{ dB}}$$

- **Step 3: invert the free space formula.**

$$\lambda = \frac{3 \times 10^8}{2.1 \times 10^9} = 0.1429 \text{ m}$$

$$L_F = 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right) \Rightarrow d = \frac{\lambda}{4\pi} 10^{L_F/20}$$

$$= \frac{0.1429}{12.566} \times 10^{117.26/20} = 0.011370 \times 10^{5.863}$$

$$= 0.011370 \times 7.2934 \times 10^5 = \boxed{8292 \text{ m} \approx 8.29 \text{ km}}$$

- **Sanity check:** substituting $d = 8.29 \text{ km}$ back gives $L_F = 32.44 + 20 \log(2100) + 20 \log(8.292) = 32.44 + 66.44 + 18.38 = 117.26 \text{ dB} \checkmark$

Problem 11 Hata model, and comparison with free space

PIN 1/23 79 Bh 4+4

QUESTION AS PRINTED 2079 Bhadra · Q3 (b) · [4+4]

In mobile propagation in a cellular system, find the correction factor and pathloss for a medium size city assuming carrier frequency as 950 MHz, height of transmitting antenna at base station is 45 m, propagation distance between antennas is 10 km and height of receiving antenna in mobile station is 5 m. Compute free space pathloss and compare it with Hata pathloss.

- **Step 1: mobile antenna correction factor**, small/medium city branch:

$$a(h_{re}) = (1.1 \log_{10} f_c - 0.7)h_{re} - (1.56 \log_{10} f_c - 0.8)$$

$$\log_{10}(950) = 2.9777$$

$$= (1.1 \times 2.9777 - 0.7)(5) - (1.56 \times 2.9777 - 0.8)$$

$$= (3.2755 - 0.7)(5) - (4.6452 - 0.8)$$

$$= 2.5755 \times 5 - 3.8452 = 12.8775 - 3.8452 = \boxed{9.032 \text{ dB}}$$

- **Step 2: the four Hata terms.** $\log_{10}(45) = 1.6532$, $\log_{10}(10) = 1$.

Term	Evaluation	Value (dB)
Constant	69.55	+69.550
$26.16 \log f_c$	26.16×2.9777	+77.897
$-13.82 \log h_{te}$	-13.82×1.6532	-22.847
$-a(h_{re})$	-9.032	-9.032
$(44.9 - 6.55 \log h_{te}) \log d$	$(44.9 - 10.828) \times 1$	+34.072
$L_{50} \text{ (urban)}$		+149.64

–

$$\boxed{L_{50}(\text{urban}) = 149.64 \text{ dB}}$$

- **Step 3: free space path loss at the same distance.**

$$PL_{FS} = 32.44 + 20 \log(950) + 20 \log(10) = 32.44 + 59.56 + 20 = 112.00 \text{ dB}$$

- **Step 4: compare.**

$$L_{50} - PL_{FS} = 149.64 - 112.00 = \boxed{37.64 \text{ dB}}$$

- **Interpretation:** the urban clutter costs almost **38 dB** over free space, ie a factor of about 5800 in power. This gap is exactly why free space is useless for cell planning and empirical models exist.
- **Validity check:** $f_c = 950 \text{ MHz}$ is inside 150–1500 MHz ✓, $d = 10 \text{ km}$ inside 1–20 km ✓, $h_{te} = 45 \text{ m}$ inside 30–200 m ✓, $h_{re} = 5 \text{ m}$ inside 1–10 m ✓. **State this; it carries a mark.**

Problem 12 Hata run backwards: maximum cell radius and area

PIN 1/23 73 Ma 6

QUESTION AS PRINTED 2073 Magh · Q3 (b) · [6]

Let us consider a medium sized city and assume the typical GSM downlink parameters. The Base Station (BS) is transmitting with power 50W. The minimum acceptable received power at Mobile Station (MS) is -91 dBm. The carrier frequency is 900 MHz, the height of BS is 30m and height of MS is 1m. Estimate the maximum cell radius and corresponding cell area using Hata Model.

The paper prints the full Hata model box in its header: $L_{50}(\text{urban}) = 69.55 + 26.16 \log f_c - 13.82 \log h_{te} - a(h_{re}) + (44.9 - 6.55 \log h_{te}) \log d$, with $a(h_{re}) = (1.1 \log f_c - 0.7)h_{re} - (1.56 \log f_c - 0.8) \text{ dB}$ for a medium sized city.

- **Step 1: allowable path loss.**

$$P_t = 10 \log_{10}(50 \times 10^3) = 46.99 \text{ dBm}$$

$$PL_{\max} = 46.99 - (-91) = 137.99 \text{ dB}$$

- **Step 2: correction factor**, medium city, $\log_{10}(900) = 2.9542$:

$$\begin{aligned} a(h_{re}) &= (1.1 \times 2.9542 - 0.7)(1) - (1.56 \times 2.9542 - 0.8) \\ &= (3.2496 - 0.7) - (4.6086 - 0.8) = 2.5496 - 3.8086 = \boxed{-1.259 \text{ dB}} \end{aligned}$$

A negative correction is normal for a 1 m mobile antenna.

- **Step 3: split Hata into** $L_{50} = A + B \log d$. With $\log_{10}(30) = 1.4771$:

$$\begin{aligned} A &= 69.55 + 26.16(2.9542) - 13.82(1.4771) - (-1.259) \\ &= 69.55 + 77.281 - 20.413 + 1.259 = 127.678 \\ B &= 44.9 - 6.55(1.4771) = 44.9 - 9.675 = 35.225 \end{aligned}$$

- **Step 4: solve for** d .

$$\begin{aligned} 137.99 &= 127.678 + 35.225 \log_{10} d \\ \log_{10} d &= \frac{137.99 - 127.678}{35.225} = \frac{10.312}{35.225} = 0.29274 \\ d &= 10^{0.29274} = \boxed{1.962 \text{ km}} \end{aligned}$$

- **Step 5: cell area.** Treating the cell as a regular hexagon of radius $R = d$:

$$A_{\text{cell}} = 2.598 R^2 = 2.598 \times (1.962)^2 = 2.598 \times 3.850 = \boxed{10.00 \text{ km}^2}$$

If the cell is taken as a circle instead, $\pi R^2 = 12.10 \text{ km}^2$. State which you used.

- **Note** $d = 1.96 \text{ km}$ is just below the Hata validity floor of 1 km, so the answer sits at the edge of the model's range. Mention this: it is worth a mark.

Problem 13 Hata in a large city

PIN 1/23 73 Bh 7

QUESTION AS PRINTED 2073 Bhadra · Q3 (b) · [7]

Determine the propagation path loss for a radio signal 900 MHz cellular system operating in a large urban city, with a base station transmitter antenna height of 100 m and mobile receiver antenna height of 2m. The mobile unit is located at a distance of 4 km. Use the Hata propagation path loss model.

Hints printed with the question: $L_{50} = 69.55 + 26.16 \log f_c - 13.82 \log h_t - \alpha(h_r) + (44.9 - 6.55 \log h_t) \log d$; $\alpha(h_r) = (1.1 \log f_c - 0.7)h_r - (1.56 \log f_c - 0.8)$ for a small to medium city, $8.29(\log 1.54h_r)^2 - 1.1$ for a large city with $f_c \leq 300 \text{ MHz}$, and $3.2(\log 11.75h_r)^2 - 4.97$ for a large city with $f_c > 300 \text{ MHz}$.

- **Step 1: pick the correct** $a(h_{re})$ **branch.** Large city, $f_c = 900 > 300 \text{ MHz}$, so

$$\begin{aligned} a(h_{re}) &= 3.2 (\log_{10}(11.75 h_{re}))^2 - 4.97 \\ &= 3.2 (\log_{10}(11.75 \times 2))^2 - 4.97 = 3.2 (\log_{10}(23.5))^2 - 4.97 \\ &= 3.2(1.3711)^2 - 4.97 = 3.2 \times 1.8799 - 4.97 = 6.0156 - 4.97 = \boxed{1.045 \text{ dB}} \end{aligned}$$

- **Step 2: the Hata terms.** $\log_{10}(900) = 2.9542$, $\log_{10}(100) = 2$, $\log_{10}(4) = 0.6021$.

Term	Evaluation	Value (dB)
Constant	69.55	+69.550
$26.16 \log f_c$	26.16×2.9542	+77.281
$-13.82 \log h_{te}$	-13.82×2	-27.640
$-a(h_{re})$	-1.045	-1.045
$(44.9 - 6.55 \log h_{te}) \log d$	$(44.9 - 13.10) \times 0.6021$	+19.147
$L_{50} \text{ (urban)}$		+137.29

$$\boxed{L_{50} = 137.29 \text{ dB}}$$

Compare with Problem 11: there the mobile was 5 m high in a medium city and the loss was 149.6 dB at 10 km. Here the BS is twice as high and the distance is 2.5 times shorter, so 137.3 dB is consistent.

Problem 14 Two-ray Vs free space path loss

PIN 1/23 70 Ma 4+4

QUESTION AS PRINTED 2070 Magh · Q3 · [4+4]

Determine the propagation path loss for signal at 800 MHz, with a transmitting antenna height of 30 m and a receiving antenna height of 2 m, over a distance of 10 km, using two-ray mobile point-to-point propagation model. How is it compared with that of free-space propagation path loss model?

- Step 1: check the model is valid.

$$\lambda = \frac{3 \times 10^8}{800 \times 10^6} = 0.375 \text{ m}$$

$$d_c = \frac{4\pi h_t h_r}{\lambda} = \frac{4\pi \times 30 \times 2}{0.375} = \frac{754.0}{0.375} = 2011 \text{ m} = 2.01 \text{ km}$$

Since $d = 10 \text{ km} > d_c$, the d^{-4} form applies. ✓ **Always do this check first: below d_c the two-ray model oscillates and this formula is wrong.**

- Step 2: two-ray path loss (unity antenna gains):

$$\begin{aligned} PL &= 40 \log_{10} d - (20 \log_{10} h_t + 20 \log_{10} h_r) \\ &= 40 \log_{10}(10\,000) - 20 \log_{10}(30) - 20 \log_{10}(2) \\ &= 40(4) - 29.54 - 6.02 = 160 - 35.56 = \boxed{124.44 \text{ dB}} \end{aligned}$$

- Step 3: free space path loss at the same distance.

$$PL_{FS} = 32.44 + 20 \log(800) + 20 \log(10) = 32.44 + 58.06 + 20 = \boxed{110.50 \text{ dB}}$$

- Step 4: comparison.

$$PL_{2\text{-ray}} - PL_{FS} = 124.44 - 110.50 = \boxed{13.94 \text{ dB worse}}$$

- **Why the two-ray model predicts more loss:** the ground-reflected ray arrives almost exactly out of phase with the direct ray and **partially cancels** it. Free space ignores that ray entirely, so it is over-optimistic.
- **Note also:** the two-ray answer contains no λ term, so it is **frequency independent**, while the free space answer is not.

Problem 15 Link budget with Okumura: reverse-link distance

PIN 1/23 76 Bh 10

QUESTION AS PRINTED 2076 Bhadra · Q3 (b) · [10]

Estimate the appropriate distance that should be maintained for reverse link between one BTS and mobile with appropriate link budget diagram.

- Mobile is connected to antenna with 20dBi gain, with a transmitting power of 15 dBm and a receive sensitivity of -75dBm .
- BTS is connected to antenna with 5 dBi gain, with a transmitting power of 20dBm and a receive sensitivity of -80dBm .
- Cables in both systems are short, with a loss of 3-dB at each side at 900 MHz frequency of operation.

Use Okumura's model for mean path loss where $G(\text{Area}) = 9\text{dB}$, $A_{mu} = 43 \text{ dB}$, $h_{re} = 10\text{m}$ and $h_{te} = 100\text{m}$. (Link margin (reverse link) = 10 dB).

- Step 1: identify the reverse link. Reverse = **mobile transmits, BTS receives**. So use the **mobile's TX power** and the **BTS's sensitivity**.
- Step 2: maximum allowable path loss on the reverse link.

Item	Value	Note
Mobile TX power	+15 dBm	
Mobile antenna gain	+20 dB	20 dBi
Mobile cable loss	−3 dB	
BTS antenna gain	+5 dB	5 dBi
BTS cable loss	−3 dB	
BTS sensitivity	−(−80) dBm	the link must land at or above this
Link margin	−10 dB	fading reserve
PL_{max}	104 dB	

$$PL_{\max} = 15 + 20 - 3 + 5 - 3 + 80 - 10 = \boxed{104 \text{ dB}}$$

- **Step 3: convert this L_{50} into a free space loss.**

$$G(h_{te}) = 20 \log\left(\frac{100}{200}\right) = -6.02 \text{ dB}, \quad G(h_{re}) = 20 \log\left(\frac{10}{3}\right) = +10.46 \text{ dB}$$

$$\begin{aligned} L_F &= L_{50} - A_{mu} + G(h_{te}) + G(h_{re}) + G_{\text{AREA}} \\ &= 104 - 43 + (-6.02) + 10.46 + 9 = 74.44 \text{ dB} \end{aligned}$$

- **Step 4: solve for d .**

$$\begin{aligned} d &= \frac{\lambda}{4\pi} 10^{L_F/20} = \frac{0.3333}{12.566} \times 10^{74.44/20} = 0.026526 \times 5271 \\ &= \boxed{139.8 \text{ m} \approx 0.14 \text{ km}} \end{aligned}$$

- **Step 5: check the forward link too.** Forward = BTS transmits, mobile receives:

$$PL_{\max} = 20 + 5 - 3 + 20 - 3 + 75 - 10 = 104 \text{ dB}$$

Identical, so the two links are perfectly balanced and the range is the same, 139.8 m. Quote both and say which limits.

- **Flag the inconsistency, it earns credit:** $A_{mu} = 43 \text{ dB}$ is Okumura's value at *tens of km*, not at 140 m. Using it at this range makes the model self-inconsistent. The honest engineering answer is that **with the stated 43 dB clutter allowance the link only closes over about 140 m**, and that either the clutter figure or the antenna gains in the question are unrealistic.

Problem 16 Feasibility of a 10 km link at 2.4 GHz

PIN 1/23 74 Bh 12 · a “prove it does not work” question, so the answer must end with a verdict

QUESTION AS PRINTED 2074 Bhadra · Q3 · [12]

Estimate the feasibility of a 10-km wireless link in suburban area, with one access point and one client radio, using Okumura model for path loss. The median attenuation value is 20 dB and gain due to environment is 13 dB. The height of access point antenna is 100 m and that of client antenna is 10 m:

- Access point is connected to antenna with 5-dBi gain, with a transmitting power of 20-dBm and a receive sensitivity of -80 -dBm
- Client is connected to antenna with 20-dBi gain, with a transmitting power of 15-dBm and a receive sensitivity of -75 -dBm
- Cables in both systems are short, with a loss of 3-dB at each side at 2.4-GHz frequency of operation.

- **Step 1: free space loss at 10 km, 2.4 GHz.**

$$\lambda = \frac{3 \times 10^8}{2.4 \times 10^9} = 0.125 \text{ m}$$

$$L_F = 32.44 + 20 \log(2400) + 20 \log(10) = 32.44 + 67.60 + 20 = 120.05 \text{ dB}$$

- **Step 2: Okumura median path loss.**

$$G(h_{te}) = 20 \log(100/200) = -6.02 \text{ dB}, \quad G(h_{re}) = 20 \log(10/3) = +10.46 \text{ dB}$$

$$L_{50} = 120.05 + 20 - (-6.02) - 10.46 - 13 = \boxed{122.61 \text{ dB}}$$

- **Step 3: forward link, AP $\rightarrow$ client.**

$$P_r = 20 + 5 - 3 + 20 - 3 - 122.61 = -83.61 \text{ dBm}$$

Client sensitivity is -75 dBm , so the margin is

$$-83.61 - (-75) = \boxed{-8.61 \text{ dB}} \quad (\text{shortfall})$$

- **Step 4: reverse link, client $\rightarrow$ AP.**

$$P_r = 15 + 20 - 3 + 5 - 3 - 122.61 = -88.61 \text{ dBm}$$

AP sensitivity is -80 dBm , so the margin is

$$-88.61 - (-80) = \boxed{-8.61 \text{ dB}} \quad (\text{shortfall})$$

- **Step 5: verdict.** Both directions fall 8.6 dB short, so the 10 km link is **NOT feasible** as specified, and it fails symmetrically, so no single end can be blamed.
- **Step 6: what distance would work?** Set the margin to zero:

$$L_{50,\max} = 122.61 - 8.61 = 114.00 \text{ dB}$$

$$L_F = 114.00 - 20 + (-6.02) + 10.46 + 13 = 111.44 \text{ dB}$$

$$d = \frac{0.125}{12.566} 10^{111.44/20} = 0.009947 \times 3.729 \times 10^5 = \boxed{3.71 \text{ km}}$$

- **How to fix it (worth stating):** raise the antenna gains, raise the AP height (each doubling of h_{te} buys 6 dB), or accept a shorter link. Raising transmit power by 8.6 dB would also do it but may breach the licence-exempt limit at 2.4 GHz.

Problem 17 Power delay profile: delay spread and coherence bandwidth

HOT 5/23 70 Ma **4+4**, 71 Ma **6**, 75 Bh **8**, 81 Bh **6** · four different papers, one method

The method, once

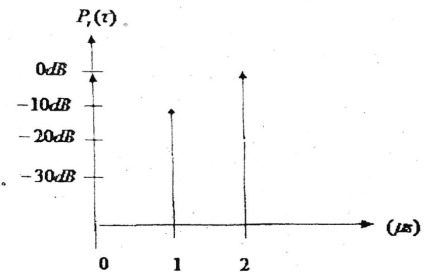
- **Step 1: convert every dB power to a linear ratio, $P_k = 10^{(\text{dB})/10}$.** **Never average in dB.**
- **Step 2: mean excess delay** $\bar{\tau} = \frac{\sum P_k \tau_k}{\sum P_k}$
- **Step 3: second moment** $\overline{\tau^2} = \frac{\sum P_k \tau_k^2}{\sum P_k}$
- **Step 4: rms delay spread** $\sigma_\tau = \sqrt{\overline{\tau^2} - \bar{\tau}^2}$
- **Step 5: coherence bandwidth** $B_c = \frac{1}{50\sigma_\tau}$ (90%) or $\frac{1}{5\sigma_\tau}$ (50%).

17.1 The 70 Ma profile (**4+4**)

QUESTION AS PRINTED 2070 Magh · Q4 · [4+4]

Define Doppler spread. Describe the types of small scale fading based on Doppler spread. Calculate the mean excess delay and rms delay spread for the multipath profile given below. Estimate the 90% and 50% coherence bandwidth of the channel.

Read the stem plot on the right carefully: it carries **three** components, 0 dB at $\tau = 0$, -10 dB at $\tau = 1 \mu\text{s}$ and 0 dB at $\tau = 2 \mu\text{s}$. **The $\tau = 0$ ray is the second arrowhead sitting on the 0 dB tick, drawn over the vertical axis. Miss it and every answer below is wrong.**



The profile as printed on the 70 Ma paper. **Two arrowheads on the vertical line: upper one is the axis, the one at 0 dB is the $\tau = 0$ ray.**

$\tau_k (\mu\text{s})$	dB	P_k	$P_k \tau_k$	$P_k \tau_k^2$
0	0	1.0	0.0	0.0
1	-10	0.1	0.1	0.1
2	0	1.0	2.0	4.0
sum		2.1	2.1	4.1

$$\bar{\tau} = \frac{2.1}{2.1} = \boxed{1.000 \mu\text{s}}$$

$$\overline{\tau^2} = \frac{4.1}{2.1} = 1.952 \mu\text{s}^2$$

$$\sigma_\tau = \sqrt{1.952 - (1.000)^2} = \sqrt{0.952} = \boxed{0.976 \mu\text{s}}$$

$$B_c(90\%) = \frac{1}{50 \times 0.976 \times 10^{-6}} = \boxed{20.5 \text{ kHz}}$$

$$B_c(50\%) = \frac{1}{5 \times 0.976 \times 10^{-6}} = \boxed{204.9 \text{ kHz}}$$

- **Maximum excess delay** within 10 dB of the peak: peak is 0 dB, and the 1 μs ray is exactly 10 dB down, so all three count and $\tau_{10\text{dB}} = 2 - 0 = 2 \mu\text{s}$.
- **Sanity check:** $\bar{\tau}$ falls on exactly 1 μs because the two equal 0 dB rays at 0 and 2 μs balance about the centre. A clean number is the sign you read the plot right.

17.2 The 75 Bh profile: greatest symbol rate without an equalizer (8)

QUESTION AS PRINTED 2075 Bhadra · Q5 · [8]

Determine the smallest symbol period T_s , and thus the greatest symbol rate that must be sent through RF channel with given power delay profile without using an equalizer.

Power [dB]	0	0	−10	−20
Delay [us]	0	50	75	100

Modulation provides suitable BER performance whenever $\sigma_\tau/T_s \leq 0.1$.

τ_k (μs)	dB	P_k	$P_k \tau_k$	$P_k \tau_k^2$
0	0	1.00	0.0	0.0
50	0	1.00	50.0	2500.0
75	−10	0.10	7.5	562.5
100	−20	0.01	1.0	100.0
sum		2.11	58.5	3162.5

$$\bar{\tau} = \frac{58.5}{2.11} = 27.725 \mu s$$

$$\overline{\tau^2} = \frac{3162.5}{2.11} = 1498.82 \mu s^2$$

$$\sigma_\tau = \sqrt{1498.82 - (27.725)^2} = \sqrt{1498.82 - 768.68} = \sqrt{730.14} = 27.02 \mu s$$

– Apply the BER condition.

$$\frac{\sigma_\tau}{T_s} \leq 0.1 \Rightarrow T_s \geq \frac{\sigma_\tau}{0.1} = 10 \sigma_\tau$$

$$T_s \geq 10 \times 27.02 \mu s = 270.2 \mu s$$

– Greatest symbol rate.

$$R_s = \frac{1}{T_s} = \frac{1}{270.2 \times 10^{-6}} = 3.70 \text{ ksymbols/s}$$

– **Interpretation:** this channel is so dispersive that without an equalizer the link is limited to under 4 ksym/s. That single number is the practical argument for equalization in Ch 5.

17.3 The 71 Ma profile: is the channel flat for AMPS and for GSM? (6)

QUESTION AS PRINTED 2071 Magh · Q3 (b) · [6]

A wireless channel is characterized by the following power-delay profile:

Power [dB]	0	−10	−20	−23
Delays [ns]	0	100	200	400

Determine the root mean square (rms) delay spread and the 90% coherence bandwidth of the above channel. Is this channel flat fading or frequency selective fading for:

- An AMPS system with transmission bandwidth 30 kHz?
- A GSM system with transmission bandwidth 200 kHz?

τ_k (ns)	dB	P_k	$P_k \tau_k$	$P_k \tau_k^2$
0	0	1.0000	0.00	0
100	−10	0.1000	10.00	1000
200	−20	0.0100	2.00	400
400	−23	0.0050	2.00	798
sum		1.1150	14.005	2201.6

$$\bar{\tau} = \frac{14.005}{1.1150} = 12.560 \text{ ns}$$

$$\overline{\tau^2} = \frac{2201.6}{1.1150} = 1974.78 \text{ ns}^2$$

$$\sigma_\tau = \sqrt{1974.78 - (12.560)^2} = \sqrt{1974.78 - 157.76} = \sqrt{1817.02} = \boxed{42.63 \text{ ns}}$$

$$B_c(90\%) = \frac{1}{50\sigma_\tau} = \frac{1}{50 \times 42.63 \times 10^{-9}} = \boxed{469.2 \text{ kHz}}$$

- (i) **AMPS, 30 kHz.** 30 kHz $\ll$ 469.2 kHz, so $B_s \ll B_c \rightarrow$ **FLAT fading**. No equalizer needed.
- (ii) **GSM, 200 kHz.** 200 kHz < 469.2 kHz, so strictly the channel is still flat, but only by a factor of **2.3**. **That is not “much less than”, so GSM is marginal on this channel** and a real GSM receiver would still run its equalizer. State it exactly this way: strictly flat, practically marginal.
- If the 50 % definition is used instead, $B_c = 1/(5\sigma_\tau) = 4.69 \text{ MHz}$ and both systems are comfortably flat. **Always say which definition you used.**

17.4 The 81 Bh profile (6)

QUESTION AS PRINTED 2081 Bhadra · Q5 · [4+6]

Compare Okumura model with Hata model. Calculate mean excess delay, rms delay spread and maximum excess delay for the multipath profile $P_r(\tau)$ given as: $P_r(\tau) = (-20\text{dB}, -10\text{dB}, -10\text{dB}, 0\text{dB})$ for $\tau = (0, 1, 2, 5)$ second. Also estimate 50% coherence bandwidth.

“second” is the paper’s own slip: the delays are microseconds. Work in μs and say so.

τ_k	dB	P_k	$P_k\tau_k$	$P_k\tau_k^2$
0	−20	0.01	0.00	0.00
1	−10	0.10	0.10	0.10
2	−10	0.10	0.20	0.40
5	0	1.00	5.00	25.00
sum		1.21	5.30	25.50

$$\bar{\tau} = \frac{5.30}{1.21} = \boxed{4.380 \text{ s}}$$

$$\overline{\tau^2} = \frac{25.50}{1.21} = 21.074 \text{ s}^2$$

$$\sigma_\tau = \sqrt{21.074 - (4.380)^2} = \sqrt{21.074 - 19.183} = \sqrt{1.891} = \boxed{1.375 \text{ s}}$$

- **Maximum excess delay (10 dB).** The peak is 0 dB at $\tau = 5$. Components within 10 dB of it are those at ≥ -10 dB, ie $\tau = 1, 2, 5$. So

$$\tau_{10\text{dB}} = 5 - 1 = \boxed{4 \text{ s}}$$

$$B_c(50\%) = \frac{1}{5 \times 1.375} = \boxed{0.1455 \text{ Hz}}$$

- **Source defect, state it:** the paper prints the delays in **seconds**. A real multipath channel has delays of **microseconds**, and a 0.15 Hz coherence bandwidth is physically impossible. **Solve it as printed**, then add one line noting that with μs the answers become $\bar{\tau} = 4.38 \mu\text{s}$, $\sigma_\tau = 1.375 \mu\text{s}$, $B_c(50\%) = 145.5 \text{ kHz}$.

Problem 18 Doppler shift and coherence time

HOT 4/23 78 Ch, 81 Ba, 80 Ch, 82 Bh

PROBLEM not a PYQ · the standard Rappaport example **verified/added**

A US digital cellular system operating at $f_c = 900 \text{ MHz}$ serves a vehicle moving at 70 km/h. Compute the received carrier frequency when the mobile moves (a) directly towards the transmitter, (b) directly away from it, and (c) perpendicular to the direction of arrival. Hence find the Doppler spread and the 50% coherence time.

Four papers ask for the Doppler *derivation* in prose (78 Ch, 81 Ba, 80 Ch, 82 Bh) and none prints numbers. This worked example is what turns that derivation into a full answer. **verified/added**

- **Step 1: convert the velocity. Always to m/s.**

$$v = 70 \frac{\text{km}}{\text{h}} \times \frac{1000}{3600} = 19.444 \text{ m/s}$$

– **Step 2: wavelength and maximum Doppler shift.**

$$\lambda = \frac{3 \times 10^8}{900 \times 10^6} = 0.3333 \text{ m}$$

$$f_m = \frac{v}{\lambda} = \frac{19.444}{0.3333} = \boxed{58.33 \text{ Hz}}$$

– **(a) Towards the transmitter, $\theta = 0^\circ$, $\cos \theta = +1$:**

$$f_d = +58.33 \text{ Hz} \Rightarrow f = 900 \text{ MHz} + 58.33 \text{ Hz} = \boxed{900.00005833 \text{ MHz}}$$

– **(b) Away, $\theta = 180^\circ$, $\cos \theta = -1$:**

$$f_d = -58.33 \text{ Hz} \Rightarrow f = \boxed{899.99994167 \text{ MHz}}$$

– **(c) Perpendicular, $\theta = 90^\circ$, $\cos \theta = 0$:**

$$f_d = 0 \Rightarrow f = \boxed{900 \text{ MHz exactly}}$$

– **Step 3: coherence time.**

$$T_c \approx \frac{0.423}{f_m} = \frac{0.423}{58.33} = 7.25 \text{ ms}$$

- **Interpretation:** the channel stays essentially constant for only 7.25 ms. A GSM burst is 0.577 ms, so the channel is safely static within one burst, which is exactly the design intent. A data frame lasting longer than 7 ms would see the channel change under it.

Ch 5 · Equalization and Diversity Numerical Problems & Solutions

0 numerical marks in the archive · **No PYQ in 23 papers has set a pure calculation on this chapter.** What is examined, repeatedly, is the **MSE optimum-weight derivation** **HOT 3/23**, which the papers ask for as bookwork with a diagram. The rest of this companion works the tutorial-set problems, because they are the standard follow-ups and they make the diversity theory concrete.

The formula sheet

- **MSE cost function** $\xi = E[x_k^2] + \omega^T \mathbf{R} \omega - 2\mathbf{p}^T \omega$
- **Optimum weights** $\hat{\omega} = \mathbf{R}^{-1} \mathbf{p}$, $\xi_{\min} = E[x_k^2] - \mathbf{p}^T \hat{\omega}$
- **LMS update** $\omega_{k+1} = \omega_k + \mu e_k \mathbf{y}_k^*$
- **Selection diversity**, M Rayleigh branches, mean SNR Γ each:

$$\Pr[\text{SNR} \leq \gamma] = \left(1 - e^{-\gamma/\Gamma}\right)^M, \quad \frac{\bar{\gamma}_M}{\Gamma} = \sum_{k=1}^M \frac{1}{k}$$

- **Maximal ratio combining** $\gamma_M = \sum_{i=1}^M \gamma_i$, so $\bar{\gamma}_M = M \Gamma$
- **Interleaver** degree m , word length n : an (n, k) code handling bursts of $b < (n-k)/2$ becomes an (mn, mk) code handling bursts of mb

Problem 1 Optimum tap weights for a 2-tap equalizer

HOT 3/23 79 Bh 8, 79 Ch 7, 80 Ch 7 · the papers ask for the **derivation**; this worked example shows the derivation actually being used

QUESTION AS PRINTED 2079 Bhadra · Q5 · [8] · 2079 Chaitra · Q6 · [7] · 2080 Chaitra · Q6 · [7]

Determine the optimal solution of the weights using MSE algorithm with its appropriate diagram and derivation.

PROBLEM not a PYQ · numbers chosen to exercise the result **verified/added**

A 2-tap adaptive equalizer sees an input correlation matrix $\mathbf{R} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$, a cross-correlation vector $\mathbf{p} = \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}$ and a transmitted signal power $E[x_k^2] = 1$. Find the optimum weight vector $\mathbf{w}_{opt}$ and the minimum mean square error ξ_{min} .

Step 1: why $\hat{\omega} = \mathbf{R}^{-1} \mathbf{p}$ (the derivation)

- Error signal, with $\mathbf{y}_k$ the input vector and x_k the known training symbol:

$$e_k = x_k - \mathbf{y}_k^T \omega$$

- Square it and take expectations:

$$\xi = E[e_k^2] = E[x_k^2] + \omega^T \mathbf{R} \omega - 2\mathbf{p}^T \omega$$

with $\mathbf{R} = E[\mathbf{y}_k \mathbf{y}_k^T]$ and $\mathbf{p} = E[x_k \mathbf{y}_k]$.

- ξ is **quadratic** in ω , so the surface is a bowl with **one** minimum. Set the gradient to zero:

$$\nabla \xi = 2\mathbf{R}\omega - 2\mathbf{p} = 0 \Rightarrow \boxed{\hat{\omega} = \mathbf{R}^{-1} \mathbf{p}}$$

Step 2: invert R

$$\det \mathbf{R} = (1)(1) - (0.5)(0.5) = 0.75$$

$$\mathbf{R}^{-1} = \frac{1}{0.75} \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} = \begin{bmatrix} 1.3333 & -0.6667 \\ -0.6667 & 1.3333 \end{bmatrix}$$

$\mathbf{R}$ must be non-singular for the solution to exist. Here $\det \mathbf{R} = 0.75 \neq 0 \checkmark$

Step 3: the optimum weights

$$\hat{\omega} = \begin{bmatrix} 1.3333 & -0.6667 \\ -0.6667 & 1.3333 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}$$

$$w_0 = 1.3333(0.8) - 0.6667(0.5) = 1.0667 - 0.3333 = 0.7333$$

$$w_1 = -0.6667(0.8) + 1.3333(0.5) = -0.5333 + 0.6667 = 0.1333$$

$$\hat{\mathbf{w}} = \begin{bmatrix} 0.7333 \\ 0.1333 \end{bmatrix}$$

Step 4: the minimum MSE

$$\xi_{\min} = E[x_k^2] - \mathbf{p}^T \hat{\mathbf{w}} = 1 - [(0.8)(0.7333) + (0.5)(0.1333)]$$

$$= 1 - (0.5867 + 0.0667) = 1 - 0.6533 = 0.3467$$

- **Interpretation:** the equalizer has removed about 65 % of the error energy. The residual 0.3467 is what no linear 2-tap filter can remove.
- **Adding taps lowers $\xi_{\min}$,** but at the cost of processing time and slower convergence. That trade-off is the practical answer to “how many taps?”.
- **The examiner wants the derivation, not the arithmetic.** Reproduce Steps 1 and 2 symbolically and show the bowl-shaped error surface; the numbers are the easy part.

Problem 2 Selection diversity: probability of a deep fade

tutorial set, and the standard companion to the “why implement diversity” question **TOP 11/23**

PROBLEM not a PYQ · the standard companion to the “why diversity” question **verified/added**

Five branches of a selection diversity receiver each receive an independent Rayleigh fading signal. The average SNR on every branch is 20 dB. Determine the probability that the SNR of the selected branch drops below 10 dB, compute the mean SNR after selection, and compare both with a single receiver having no diversity.

“Why is there a need to implement diversity?” is asked in 11 of the 23 papers; this is the arithmetic that answers it with a number. **verified/added**

Step 1: form the ratio γ/Γ (in linear terms, not dB)

- Threshold $\gamma = 10 \text{ dB} = 10^{10/10} = 10$
- Mean per branch $\Gamma = 20 \text{ dB} = 10^{20/10} = 100$
-

$$\frac{\gamma}{\Gamma} = \frac{10}{100} = 0.1$$

Subtracting in dB gives the same answer, but watch the order: $10 \text{ dB} - 20 \text{ dB} = -10 \text{ dB}$, and $10^{-10/10} = 0.1$. ✓ Either route is fine; mixing them is not.

Step 2: probability the SNR falls below the threshold

$$\Pr[\text{SNR} \leq \gamma] = \left(1 - e^{-\gamma/\Gamma}\right)^M = \left(1 - e^{-0.1}\right)^M = (0.09516)^M$$

Branches M	$\Pr[\text{SNR} \leq 10 \text{ dB}]$	Meaning
1 (no diversity)	9.516×10^{-2}	the link is below threshold 9.5 % of the time
2	9.056×10^{-3}	about 1 in 110
4	8.201×10^{-5}	about 1 in 12 000
5	7.804×10^{-6}	about 1 in 128 000

- **Five-branch selection diversity improves the outage probability by more than four orders of magnitude** over a single receiver, from 9.5×10^{-2} to 7.8×10^{-6} .
- That is the entire justification for diversity, expressed as a number.

Step 3: mean SNR after selection combining

$$\frac{\bar{\gamma}_M}{\Gamma} = \sum_{k=1}^M \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = 2.2833$$

$$\bar{\gamma}_5 = 100 \times 2.2833 = 228.33 = 23.59 \text{ dB}$$

- So the mean SNR rises only **3.59 dB**, from 20 dB to 23.59 dB.
- **The important point:** the gain in **mean SNR** is modest, but the gain in **outage probability** is enormous. Diversity works by cutting the **tail** of the distribution, not by lifting the average. **Say this in the theory answer too.**
- **Diminishing returns:** $M=2$ gives 1.5Γ (1.76 dB), $M=3$ gives 1.833Γ (2.63 dB), $M=4$ gives 2.083Γ (3.19 dB), $M=5$ gives 2.283Γ (3.59 dB). **Most of the benefit is already in the second branch**, which is why practical systems use 2, occasionally 4, and almost never 5.

Step 4: compare with maximal ratio combining

- MRC output SNR is the **sum** of the branch SNRs, so

$$\bar{\gamma}_M = M\Gamma = 5 \times 100 = 500 = \boxed{26.99 \text{ dB}}$$

- MRC gains a full **6.99 dB** against selection diversity's 3.59 dB, and its outage probability at 10 dB falls to 7.67×10^{-8} , another factor of 100 better.
- **The price** is M full receivers plus accurate estimation of each branch's amplitude and phase. Selection diversity needs only to compare SNRs.

Problem 3 Interleaver design against burst errors

HOT 3/23 80 Bh, 81 Bh, 81 Ba · the papers ask for the **principle**; this shows the design sum behind it

QUESTION AS PRINTED 2080 Bhadra · Q5 (b) · [4] · 2081 Bhadra · Q7 · [4] · 2081 Baishakh · Q5 · [2+6]

Why interleaving is needed in wireless communication? Explain its working in brief. · Elaborate the working principle behind Interleaving with an appropriate diagram. · Define Interleaving.

PROBLEM not a PYQ · the design sum behind the principle **verified/added**

An interleaver has m rows of n -bit words. Each word carries k source bits and $(n - k)$ parity bits of a block code. Show how the interleaver and coder together break up a burst of channel errors, and size the interleaver for a $(15, 11)$ block code with interleaver degree $m = 8$: give the block size, the source bits per block, and the longest burst the arrangement can correct.

Step 1: what the block code alone can do

- A block code with minimum distance d' corrects t errors when $d' \geq 2t + 1$.
- For an (n, k) code the parity budget is $n - k$, so it can handle a burst of

$$b < \frac{n - k}{2}$$

- For the $(15, 11)$ code: $n - k = 4$, so $t = 2$ and a burst of up to **2 bits** per codeword is correctable.

Step 2: what the interleaver does

- Coded bits are written into the array **by column** and read out **by row**.
- Two bits that were adjacent in a codeword end up m **bit periods apart** on the channel.
- So a channel burst of length $l = mb$ is spread across m different codewords, giving each of them only b errors, which each can correct.

Step 3: the resulting combined code

- An (n, k) code plus a degree- m interleaver behaves as an

$$(mn, mk) \text{ block code}$$

- For $(15, 11)$ with $m = 8$:

$$(8 \times 15, 8 \times 11) = \boxed{(120, 88) \text{ block code}}$$

- Burst-handling capability:

$$l = m \times b = 8 \times 2 = \boxed{16 \text{ consecutive bits}}$$

- **The code rate is unchanged** at $88/120 = 11/15 = 0.733$. The interleaver costs no redundancy at all: it buys burst protection purely with **delay and memory**.

Step 4: the delay penalty

- The interleaver must fill all $mn = 120$ bits before the first row can be sent, and the de-interleaver must fill 120 bits before the first row can be decoded.
- At a coded rate of, say, 22.8 kbps (GSM full rate) that is

$$\frac{2 \times 120}{22\,800} = 10.5 \text{ ms of added delay}$$

- **Human speech tolerates delays below about 40 ms**, which is the hard cap on how deeply a voice system may interleave. Data services, with no such limit, interleave far more deeply.

Problem 4 Choosing the equalizer algorithm from the channel

the reasoning behind “explain any two adaptive equalization algorithms” (73 Ma, 74 Ma)

QUESTION AS PRINTED · 2073 Magh · Q5 (a) · [4] · 2074 Magh · Q10 (a) · [5]

Explain briefly adaptive equalization algorithms (any two). · Write short note on Adaptive Equalization.

PROBLEM not a PYQ · the reasoning that picks the algorithm verified/added

A GSM mobile travels at 70 km/h on a 900 MHz carrier through a channel whose rms delay spread is $\sigma_\tau = 5 \mu\text{s}$. The symbol rate is 270.833 ksym/s. Decide how many equalizer taps are needed, how often the equalizer must retrain, and hence which adaptive algorithm (LMS, RLS or Kalman) suits the link.

- **Step 1: how many taps?** The tap span must cover the **maximum expected delay spread**, in symbol periods.

$$T_s = \frac{1}{270\,833} = 3.692 \mu\text{s}$$

$$\frac{\sigma_\tau}{T_s} = \frac{5}{3.692} = 1.35 \text{ symbols}$$

Allowing for the maximum excess delay being several times σ_τ , an equalizer spanning **about 5 symbol periods** is needed. **GSM equalizers are specified to handle 4 to 5 symbols of delay spread, which matches.**

- **Step 2: how fast does the channel change?**

$$v = 70 \text{ km/h} = 19.44 \text{ m/s}, \quad \lambda = 0.3333 \text{ m}, \quad f_m = \frac{v}{\lambda} = 58.33 \text{ Hz}$$

$$T_c \approx \frac{0.423}{f_m} = \frac{0.423}{58.33} = 7.25 \text{ ms}$$

- **Step 3: retraining rate.** The equalizer must converge and be used well inside 7.25 ms.

$$\text{minimum retrain rate} = \frac{1}{7.25 \text{ ms}} = 138 \text{ times per second}$$

GSM sends a training midamble in every 4.615 ms frame, ie 217 times per second, comfortably inside the requirement. ✓

- **Step 4: which algorithm?**

- The channel changes every few milliseconds, and the GSM training sequence is only **26 bits** long.
- **LMS** typically needs hundreds of iterations to converge, so 26 bits is not enough.
- **RLS** converges in roughly $2N$ iterations for an N -tap filter, ie about 10 for a 5-tap equalizer. **RLS, or an MLSE with a channel estimator, is the only viable choice.**
- That is exactly why GSM receivers use **MLSE with Viterbi**, not a simple LMS transversal equalizer.

Problem 5 Cell-edge SNR gain from two-branch diversity

the arithmetic behind “how diversity improves the quality of the network” (76 Bh 2)

QUESTION AS PRINTED 2076 Bhadra · Q5 · [2+4]

Explain how diversity improves the quality of network. Explain the operation of RAKE receiver with appropriate block diagram.

PROBLEM not a PYQ · the number that answers “how much” verified/added

On a Rayleigh channel the mean SNR at the cell edge is 15 dB and the receiver needs 8 dB to demodulate. Compare the outage probability with a single antenna and with two-branch selection diversity, and state the coverage gain this implies.

– **Step 1: the ratio.**

$$\gamma = 10^{8/10} = 6.31, \quad \Gamma = 10^{15/10} = 31.62, \quad \frac{\gamma}{\Gamma} = 0.1995$$

– **Step 2: outage with one antenna.**

$$\text{Pr} = 1 - e^{-0.1995} = 0.1809 = \boxed{18.1 \%}$$

– **Step 3: outage with two branches.**

$$\text{Pr} = (0.1809)^2 = 0.03273 = \boxed{3.27 \%}$$

– **Step 4: what that is worth in dB.** To reach 3.27 % outage with a *single* antenna the mean SNR would have to satisfy $1 - e^{-\gamma/\Gamma'} = 0.03273$, ie $\gamma/\Gamma' = 0.03327$, giving

$$\Gamma' = \frac{6.31}{0.03328} = 189.6 = 22.78 \text{ dB}$$

$$\text{diversity gain} = 22.78 - 15 = \boxed{7.78 \text{ dB}}$$

– **Interpretation:** a second antenna is worth almost 8 dB of transmit power at this operating point, for no extra spectrum and no extra power. **That is why every base station has two receive antennas.**

– **Coverage implication.** With a path loss exponent $n = 3.5$, 7.78 dB of margin extends the cell radius by

$$10^{7.78/(10 \times 3.5)} = 10^{0.2223} = 1.67 \times$$

ie **about 2.8 times the area from the same site.**

Ch 6 · Speech and Channel Coding Numerical Problems & Solutions

5 numerical marks, 0.2% of the archive, 2% of this chapter · The examined calculation here is the **7-bit Hamming code** (80 Ch 5), worked both ways below. Convolutional encoding, Viterbi decoding **HOT 5/23** are asked as "explain", but they are only convincing with a worked trellis, so one is given. The remaining problems are the tutorial-set coder-rate calculations, which are the standard follow-ups to the speech coding theory.

The formula sheet

- **Hamming code** $n = 2^m - 1$, $k = 2^m - m - 1$, $n - k = m$, $d_{\min} = 3$, $t = 1$, $\text{rate} = k/n$
- **Detect** t errors: $d' \geq t + 1$. **Correct** t errors: $d' \geq 2t + 1$
- **Convolutional code** $\text{rate} = k/n$, constraint length K , 2^{K-1} trellis states
- **Nyquist** a band-pass signal of bandwidth B needs $f_s = 2B$ samples/s
- **Bit rate** $R = f_s \times$ (bits per sample)
- **Quantizer** $D = \int (x - f_Q(x))^2 p(x) dx$, $S = E[x^2] = \int x^2 p(x) dx$, $\text{SDR} = 10 \log_{10}(S/D)$
- **Coding gain** $G = \left(\frac{E_b}{N_0} \right)_{\text{uncoded}} - \left(\frac{E_b}{N_0} \right)_{\text{coded}}$ dB, at the same BER

Problem 1 7-bit Hamming code: encode and then detect

PIN 1/23 80 Ch 5 · **the only pure channel-coding calculation in 23 papers, so it is worth banking exactly**

QUESTION AS PRINTED 2080 Chaitra · Q7 · [3+5]

Explain the characteristics of Speech. Explain 7-bit hamming code method with relevant example.

The paper prints no data word, so supply your own. The run below encodes **1011** into a (7, 4) Hamming codeword, then receives a corrupted word and locates the faulty bit: that is the "relevant example" the 5 marks are for. **verified/added**

Step 0: the structure, once

- For $m = 3$: $n = 2^3 - 1 = 7$, $k = 2^3 - 3 - 1 = 4$, parity bits = 3.
- Parity bits go in positions that are **powers of 2**: 1, 2, 4. Data fills the rest: 3, 5, 6, 7.
- Layout, written with bit 7 on the *left* and bit 1 on the *right*:

$$D_7 \ D_6 \ D_5 \ P_4 \ D_3 \ P_2 \ P_1$$

- Each parity bit checks the positions whose binary index contains its own bit:

Parity bit	Checks positions	Binary indices	Why
P_1	1, 3, 5, 7	001, 011, 101, 111	all have bit 0 set
P_2	2, 3, 6, 7	010, 011, 110, 111	all have bit 1 set
P_4	4, 5, 6, 7	100, 101, 110, 111	all have bit 2 set

This is the whole trick. Because the groups are built on the binary indices, the three parity checks read together **are** the binary address of the faulty bit.

Part A Encode the data word 1011 with even parity

- **Step 1: place the data bits into positions 7, 6, 5, 3:**

$$D_7 = 1, \quad D_6 = 0, \quad D_5 = 1, \quad D_3 = 1$$

- **Step 2: compute each parity bit so that its group has an even number of ones.**

Parity	Group contents	Ones already present	Set to
P_1	$(P_1, D_3, D_5, D_7) = (P_1, 1, 1, 1)$	3 (odd)	1
P_2	$(P_2, D_3, D_6, D_7) = (P_2, 1, 0, 1)$	2 (even)	0
P_4	$(P_4, D_5, D_6, D_7) = (P_4, 1, 0, 1)$	2 (even)	0

- **Step 3: assemble the codeword, bit 7 first:**

Position	7	6	5	4	3	2	1
Contents	D_7	D_6	D_5	P_4	D_3	P_2	P_1
Value	1	0	1	0	1	0	1

transmitted codeword = 1010101

Part B A codeword 1011011 is received. Is it correct? If not, locate the error.

- Step 1: lay out the received bits.

Position	7	6	5	4	3	2	1
Received	1	0	1	1	0	1	1

- Step 2: recheck each parity group, still assuming even parity.

Check	Positions	Bits read	Parity	Syndrome bit
P_1	1, 3, 5, 7	1, 0, 1, 1	ODD → error	1
P_2	2, 3, 6, 7	1, 0, 0, 1	EVEN → ok	0
P_4	4, 5, 6, 7	1, 1, 0, 1	ODD → error	1

- Step 3: read the syndrome. Write it as $P_4P_2P_1$:

$$\text{syndrome} = 101_{(2)} = 4 + 0 + 1 = 5$$

- Step 4: conclude.

The received word is INCORRECT; bit position 5 is in error.

- Step 5: correct it by flipping bit 5, $1 \rightarrow 0$:

$$\text{corrected codeword} = 1001011$$

$$\text{recovered data (positions 7,6,5,3)} = 1000$$

- **Non-zero syndrome = error, and its decimal value is the position. A zero syndrome means no single-bit error.**
- **Limitation to state:** $d_{\min} = 3$, so this code corrects **one** bit and detects **two**. A **two-bit** error produces a non-zero syndrome pointing at the *wrong* position, and “correcting” it makes things worse. **That is why Hamming codes are always paired with interleaving** (Ch 5) on a fading channel, where errors arrive in bursts.

Hamming code parameters for other m

m	$n = 2^m - 1$	$k = 2^m - m - 1$	parity bits	rate k/n
3	7	4	3	0.571
4	15	11	4	0.733
5	31	26	5	0.839
6	63	57	6	0.905

The rate improves as m grows, but the code still corrects only **one** bit per block, so the protection per bit gets *weaker*. That trade-off is worth one line in any Hamming answer.

Problem 2 Convolutional encoding and Viterbi decoding

HOT 5/23 74 Bh **5**, 75 Bh, 70 Ma, 72 Ash **3**, 73 Bh **5**

QUESTION AS PRINTED 5 papers, always as a short note

Write short notes on: **Convolutional encoding and decoding** (2074 Bhadra Q9 c **5**)

Write short notes on: **Viterbi decoding algorithm** (2073 Bhadra Q9 c **5** · 2072 Ashwin Q9 c **3** · 2075 Bhadra Q12 b · 2070 Magh Q10 c)

PROBLEM not a PYQ · the example that carries the short note **verified/added**

For the rate $1/2$, constraint length $K = 3$ convolutional encoder defined by $X_1 = M + M_1 + M_2$ and $X_2 = M + M_2$ (mod-2 addition), encode the message **1011**. Then decode the received sequence $Y = 11\ 01\ 10$ using the Viterbi algorithm, showing the trellis and the surviving path metrics.

Part A Encoding 1011

- State = the contents of the two memory stages (M_1M_2), so there are $2^{K-1} = 4$ states: $a = 00$, $b = 01$, $c = 10$, $d = 11$.
- Two flush (tail) bits are appended so the encoder returns to state a .

Input M	State (M_1M_2)	$X_1 = M + M_1 + M_2$	$X_2 = M + M_2$	Output	Next state
1	00	$1+0+0 = 1$	$1+0 = 1$	11	10
0	10	$0+1+0 = 1$	$0+0 = 0$	10	01
1	01	$1+0+1 = 0$	$1+1 = 0$	00	10
1	10	$1+1+0 = 0$	$1+0 = 1$	01	11
0	11	$0+1+1 = 0$	$0+1 = 1$	01	01
0	01	$0+0+1 = 1$	$0+1 = 1$	11	00

encoded output = 11 10 00 01 01 11

- 4 message bits plus 2 flush bits give 6 input bits, and at rate $1/2$ that is **12 output bits**. ✓
- Code rate including the tail** is $4/12 = 1/3$, not $1/2$. Real systems use long frames so the tail overhead is negligible. Worth a line.

Part B Viterbi decoding of $Y = 11\ 01\ 10$

- Metric used: Hamming distance** between the branch's expected output and the received pair. Lower is better.
- At every node keep only the **smaller-metric** incoming path (the survivor) and discard the other.

Step	Received	State	Path metric	Surviving input path
1	11	00	2	0
		10	0	1
2	01	00	3	0, 0
		01	2	1, 0
		10	3	0, 1
		11	0	1, 1
3	10	00	3	1, 0, 0
		01	2	1, 1, 0
		10	3	1, 0, 1
		11	0	1, 1, 1

Reading the table

- Step 1.** From state 00, input 0 gives output 00 (distance 2 from the received 11), input 1 gives output 11 (distance 0). So state 10 carries metric 0.
- Step 2.** From state 10, input 0 gives 10 (distance 1 from 01) and input 1 gives 01 (distance 0). The path 1, 1 into state 11 keeps metric 0.
- Step 3.** From state 11, input 1 gives output 10, which **matches the received 10 exactly**, so the metric stays 0.
- Survivor:** the path ending in state 11 with total metric 0.

decoded message = 1 1 1

- Verify by re-encoding** 1, 1, 1 from state 00: 11, then 01, then 10 = Y exactly. ✓ **Metric 0 means the received sequence was a valid codeword, ie no channel errors.**

- **Why Viterbi is tractable:** after every step there are still only 4 survivors, never more, because each state keeps one path. An exhaustive tree search would have 2^L paths after L bits.

Problem 3 Sub-band coder bit rate

PIN 1/23 80 Bh 6 · the calculation behind “sub-band coding with transmitter and receiver”

QUESTION AS PRINTED 2080 Bhadra · Q6 (a) · [6]

Explain sub-band coding with transmitter and receiver.

PROBLEM not a PYQ · the rate sum behind the block diagram verified/added

A sub-band coder splits the input into the four sub-bands tabulated below and samples each at its band-pass Nyquist rate. Find the total encoding bit rate, and compare it with coding the full band at the same resolution.

Sub-band	Band (Hz)	Bandwidth	$f_s = 2B$	Bits/sample	Rate (bps)
1	225–450	225 Hz	450	4	1800
2	450–900	450 Hz	900	3	2700
3	1000–1500	500 Hz	1000	2	2000
4	1800–2700	900 Hz	1800	1	1800
Total					8300

- **Step 1: bandwidth of each band** $= f_{\text{high}} - f_{\text{low}}$.
- **Step 2: sampling rate.** For **perfect reconstruction** each band-pass signal must be sampled at **twice its own bandwidth**, not twice its highest frequency. **That saving is the whole reason for the low-pass translation step in the sub-band coder.**

$$f_{s,1} = 2(450 - 225) = 450 \text{ samples/s}, \quad f_{s,2} = 2(900 - 450) = 900$$

$$f_{s,3} = 2(1500 - 1000) = 1000, \quad f_{s,4} = 2(2700 - 1800) = 1800$$

- **Step 3: total rate.**

$$R = 450(4) + 900(3) + 1000(2) + 1800(1)$$

$$= 1800 + 2700 + 2000 + 1800 = \boxed{8300 \text{ bps} = 8.3 \text{ kbps}}$$

- **Note the bit allocation:** the **narrowest, lowest** band gets the **most** bits (4), and the widest, highest band gets the fewest (1). That follows directly from the **PSD property** of speech: low bands carry most of the energy and most of the perceptual weight.
- **Compare:** plain 8 kHz / 8-bit PCM of the same speech would need **64 kbps**. Sub-band coding achieves nearly **8:1 compression** while remaining a waveform coder.

Problem 4 Upper bound on speech coder rate in an FDMA system

tutorial set; the calculation behind “choosing a speech codec for mobile communications”

PROBLEM not a PYQ · the sum behind “choosing a speech codec” verified/added

A digital mobile system uses 810–826 MHz for the forward channel and 940–956 MHz for the reverse channel. 90% of the bandwidth is given to traffic channels, and the system must support at least 1150 simultaneous calls using FDMA. The modulation achieves a spectral efficiency of 1.68 bps/Hz, and channel impairments force rate 1/2 forward error correction. Find the upper bound on the transmission bit rate that a speech coder in this system can use.

- **Step 1: total band.**

$$B_{\text{total}} = 826 - 810 = 16 \text{ MHz}$$

The reverse band is the same width, and FDMA pairs them, so work with one direction.

- **Step 2: bandwidth actually available to traffic.**

$$B_{\text{traffic}} = 0.9 \times 16 = 14.4 \text{ MHz}$$

The other 10 % goes to control channels and guard bands.

- **Step 3: bandwidth per channel.** FDMA divides it among the simultaneous calls:

$$B_{\text{ch}} = \frac{14.4 \text{ MHz}}{1150} = 12.52 \text{ kHz}$$

$$\Rightarrow \boxed{\text{maximum channel bandwidth} = 12.5 \text{ kHz}}$$

- **Step 4: raw bit rate the channel can carry.**

$$R_{\text{raw}} = 1.68 \frac{\text{bps}}{\text{Hz}} \times 12.52 \text{ kHz} = 21.04 \text{ kbps}$$

- **Step 5: subtract the FEC overhead.** A rate 1/2 code doubles the bits, so only half the raw rate is available to the speech coder:

$$R_{\text{speech}} = 0.5 \times 21.04 = \boxed{10.52 \text{ kbps}}$$

- **Interpretation:** the speech coder must run at **10.5 kbps or below**. A plain waveform coder cannot do that at toll quality, so this system needs a **hybrid or vocoder** class coder, eg RPE-LTP at 13 kbps (too fast here) or a CELP variant at 8 kbps (fits).
- **This is exactly the reasoning that forces cellular systems onto low-rate coders**, and it is the numerical answer to “why do we need speech coding techniques”.

Problem 5 Quantizer mean square error and signal to distortion ratio

tutorial set; the calculation behind the PDF property of speech **TOP 7/23**

PROBLEM not a PYQ · the PDF property of speech, made numerical **verified/added**

The input signal to a quantizer has the probability density function $p(x) = x/32$ for $0 \leq x \leq 8$ and zero elsewhere. The quantization levels are $\{1, 3, 5, 7\}$.

- Compute the mean square error distortion at the quantizer output.
- Compute the output signal to distortion ratio.
- How would you redistribute the quantization levels to decrease the distortion, and for what input pdf would this quantizer already be optimal?

- **Step 0: check the pdf is valid.**

$$\int_0^8 \frac{x}{32} dx = \left[\frac{x^2}{64} \right]_0^8 = \frac{64}{64} = 1 \quad \checkmark$$

- **Step 1: set the decision boundaries.** With levels at 1, 3, 5, 7 the midpoints give boundaries at $\{0, 2, 4, 6, 8\}$, so the quantizer is **uniform** with step 2.
- **Step 2: mean square error distortion.**

$$\begin{aligned} D &= \int_{-\infty}^{\infty} (x - f_Q(x))^2 p(x) dx \\ &= \int_0^2 (x - 1)^2 \frac{x}{32} dx + \int_2^4 (x - 3)^2 \frac{x}{32} dx + \int_4^6 (x - 5)^2 \frac{x}{32} dx + \int_6^8 (x - 7)^2 \frac{x}{32} dx \end{aligned}$$

- **Step 3: evaluate one integral to show the method.** For the first,

$$\begin{aligned} \int_0^2 (x^2 - 2x + 1) \frac{x}{32} dx &= \frac{1}{32} \int_0^2 (x^3 - 2x^2 + x) dx \\ &= \frac{1}{32} \left[\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} \right]_0^2 = \frac{1}{32} \left(4 - \frac{16}{3} + 2 \right) = \frac{1}{32} \cdot \frac{2}{3} = \frac{1}{48} \end{aligned}$$

The other three evaluate to $\frac{3}{48}$, $\frac{5}{48}$ and $\frac{7}{48}$ by the same method.

- **Step 4: total distortion.**

$$D = \frac{1 + 3 + 5 + 7}{48} = \frac{16}{48} = \frac{1}{3} = \boxed{0.3333}$$

- **Step 5: signal power.**

$$S = E[x^2] = \int_0^8 x^2 \frac{x}{32} dx = \frac{1}{32} \left[\frac{x^4}{4} \right]_0^8 = \frac{1}{32} \cdot \frac{4096}{4} = \boxed{32}$$

– **Step 6: signal to distortion ratio.**

$$\text{SDR} = 10 \log_{10} \left(\frac{S}{D} \right) = 10 \log_{10} \left(\frac{32}{0.3333} \right) = 10 \log_{10}(96.0) = \boxed{19.82 \text{ dB}}$$

– **Step 7: how to reduce the distortion.**

- Note that the four sub-integrals came out as 1, 3, 5, 7 over 48: **the top interval contributes seven times as much distortion as the bottom one.**
- That is because $p(x) = x/32$ rises with x , so most of the probability mass sits at large amplitudes.
- **Fix: use a non-uniform quantizer.** Place **more levels where the probability is high** (large x) and fewer where it is low (small x), ie shrink the step size towards $x = 8$.
- This is exactly the **PDF property** of speech being exploited, and it is the principle behind μ -law and A-law companding.
- **Step 8: for what pdf is this quantizer optimal?** A **uniform** quantizer is optimal when every interval contributes equally, which happens only when the input is **uniformly distributed** over $[0, 8]$, ie $p(x) = 1/8$.

Problem 6 Speech coder rates: what fits in what

supporting numbers for the vocoder **TOP 10/23** and LPC **TOP 8/23** questions

Reference figures. **Quoting one correct rate is often worth a mark** in an otherwise descriptive answer.

Coder	Class	Bit rate	Note
PCM (toll reference)	Waveform	64 kbps	8 kHz, 8 bit, μ -law
ADPCM	Waveform	32 kbps	G.726
Sub-band coding	Waveform	9.6–32 kbps	4 or 8 bands
Adaptive transform (ATC)	Waveform	9.6–20 kbps	DCT based
Voice-excited vocoder	Hybrid	7.2–9.6 kbps	no pitch extraction needed
GSM RPE-LTP	Hybrid	13 kbps	full rate GSM
CELP	Vocoder	4.8 kbps	codebook excitation
LPC	Vocoder	4.8 kbps (good quality), 2.4 kbps	all-pole model

- **Compression achieved** against the 64 kbps PCM reference: ADPCM $2\times$, sub-band $\approx 7\times$, GSM $\approx 5\times$, CELP $\approx 13\times$.
- **The pattern to remember:** rate falls, complexity rises, and robustness falls. A 4.8 kbps CELP frame damaged by the channel is far more audible than a damaged 64 kbps PCM sample. **That is why the lowest-rate coders need the strongest channel coding**, which is the link between the two halves of this chapter.

Problem 7 Coding gain

syllabus, no PYQ yet · included because it is the number that justifies every code in the chapter

PROBLEM not a PYQ · the number that justifies every code here **verified/added**

An uncoded BPSK link needs $E_b/N_0 = 10.5$ dB to reach a BER of 10^{-5} . With a rate $1/2$, $K = 7$ convolutional code and soft-decision Viterbi decoding, the same BER is reached at $E_b/N_0 = 4.5$ dB. Find the coding gain in dB and the transmit power saving it represents.

– **Step 1: coding gain.**

$$G = \left(\frac{E_b}{N_0} \right)_{\text{uncoded}} - \left(\frac{E_b}{N_0} \right)_{\text{coded}} = 10.5 - 4.5 = \boxed{6 \text{ dB}}$$

– **Step 2: what 6 dB is worth in power.**

$$10^{6/10} = 3.98 \approx 4$$

The transmitter can run at one quarter of the power for the same quality.

– **Step 3: what it is worth in range**, at path loss exponent $n = 3.5$:

$$10^{6/(10 \times 3.5)} = 10^{0.1714} = 1.48$$

ie **48 % more range**, or $1.48^2 = 2.2$ times the area from one site.

- **Step 4: what it costs.** The rate $1/2$ code **doubles the transmitted bit rate**, so it needs twice the bandwidth (or halves the user data rate).
- **TCM exists precisely to escape that cost:** by combining coding with a larger signal constellation it delivers coding gains up to 6 dB with **no bandwidth expansion**. Turbo codes then push the gain further still, close to the Shannon bound.

Ch 7 · Multiple Access Numerical Problems & Solutions

72 numerical marks, 3.5% of the archive, 23% of this chapter · Every problem below is a real PYQ, with its year tags and marks, except the four marked **PROBLEM**, which are tutorial or textbook problems added to cover a pattern the papers imply but never set. Solutions are worked in full: what was calculated, why, and which formula was used. **Nothing is skipped to save space.**

The four formulas everything here is built on

- **FDMA channels** $N = \frac{B_t - 2B_{\text{guard}}}{B_c}$
- **TDMA channels** $N = \frac{m(B_{\text{tot}} - 2B_{\text{guard}})}{B_c}$, $m = \text{slots per carrier}$
- **TDMA frame efficiency** $\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\%$
- **Hadamard recursion** $H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix}$, $H_1 = [1]$

B_t = total spectrum allocation, B_{guard} = guard band at one edge, B_c = channel (carrier) bandwidth, b_{OH} = overhead bits per frame, b_T = total bits per frame. **All three bandwidths must be in the same unit before you divide.** Convert everything to Hz, or everything to kHz, and never mix.

Problem 1 Number of channels available in an FDMA system

HOT 4/23 80 Ba **3+2**, 79 Ch **4**, 77 Ch **4**, 78 Ch **4** · **same formula three times**, only the numbers change

1.1 $B_t = 25 \text{ MHz}$, **guard** = 100 kHz, **channel** = 200 kHz (80 Ba)

QUESTION AS PRINTED 2080 Baishakh · Q7(a) · [3+2]

Explain the non-linear effect in FDMA. If the total spectrum allocation is 25 MHz, the guard band allocated at the edge of the spectrum is 100 KHz, and the channel bandwidth is 200 KHz. Find the number of channels available in an FDMA system.

The 3-mark theory half is answered in the chapter, §7.2. Only the 2-mark calculation is worked here.

- **Step 1: list the given data and put it all in one unit.** The formula divides bandwidths by each other, so a mixed unit silently gives an answer 1000 times wrong.
 - $B_t = 25 \text{ MHz} = 25 \times 10^6 \text{ Hz}$
 - $B_{\text{guard}} = 100 \text{ kHz} = 100 \times 10^3 = 0.1 \times 10^6 \text{ Hz}$
 - $B_c = 200 \text{ kHz} = 200 \times 10^3 = 0.2 \times 10^6 \text{ Hz}$
- **Step 2: subtract the guard bands.** **A guard band sits at each edge of the allocation**, at the bottom and at the top, so $2B_{\text{guard}}$ is lost before any channel is cut. This factor of 2 is the single most common slip in this problem.

$$B_t - 2B_{\text{guard}} = 25 \times 10^6 - 2(100 \times 10^3) = 25,000,000 - 200,000 = 24,800,000 \text{ Hz}$$

- **Step 3: divide the usable spectrum by the channel bandwidth.**

$$N = \frac{B_t - 2B_{\text{guard}}}{B_c} = \frac{24,800,000}{200 \times 10^3} = \frac{24,800}{200} = 124$$

- **Step 4: state the answer as a whole number.** N counts channels, so it is an integer. Here the division is exact, and where it is not, **round down**: a fractional channel cannot be assigned.
- **Final answer: $N = 124$ channels.**

1.2 $B_t = 12.8 \text{ MHz}$, **guard** = 10 kHz, **channel** = 30 kHz (79 Ch, 77 Ch)

QUESTION AS PRINTED 2079 Chaitra · Q8(b) and 2077 Chaitra · Q10(b) · [4]

Calculate the number of radio channels available in the FDMA system with following data. A US analog mobile phone system is allocated 12.8 MHz fir every simplex band. The total spectrum allocated is 12.8 MHz, the guard bandwidth is 10 KHz and the channel bandwidth is 30 KHz.

“fir” is the 2079 Chaitra paper’s own typo for “for”; 2077 Chaitra prints the same question correctly. The two papers are otherwise word for word identical.

- **Step 1: read the statement carefully.** It says 12.8 MHz twice: once as “allocated for every simplex band” and once as “the total spectrum allocated”. **These are the same number, not two numbers to add.** $B_t = 12.8$ MHz is what goes into the formula.
 - The phrase “every simplex band” is telling you this is **one direction only**. A full duplex system would need a second 12.8 MHz for the reverse link, and the channel count would still be 12.8 MHz worth.
- **Step 2: convert to one unit.** Working in kHz keeps the numbers small: $B_t = 12,800$ kHz, $B_{\text{guard}} = 10$ kHz, $B_c = 30$ kHz.
- **Step 3: remove both guard bands.**

$$B_t - 2B_{\text{guard}} = 12,800 - 2(10) = 12,800 - 20 = 12,780 \text{ kHz}$$

- **Step 4: divide by the channel bandwidth.**

$$N = \frac{12,780}{30} = 426$$

- **Final answer: $N = 426$ radio channels.**

1.3 $B_t = 12.5$ MHz, guard = 10 kHz, channel = 30 kHz (78 Ch)

QUESTION AS PRINTED 2078 Chaitra · Q7(b) · [4]

If total bandwidth is 12.5 MHz, guard bandwidth is 10 KHz and channel bandwidth is 30 KHz, find the number of channels available in an FDMA system.

- **Step 1: same formula, in kHz.** $B_t = 12,500$ kHz, $B_{\text{guard}} = 10$ kHz, $B_c = 30$ kHz.
- **Step 2: remove both guard bands.**

$$B_t - 2B_{\text{guard}} = 12,500 - 20 = 12,480 \text{ kHz}$$

- **Step 3: divide.**

$$N = \frac{12,480}{30} = 416$$

- **Final answer: $N = 416$ channels.**
- **Worth knowing:** 416 is the real AMPS figure. In the US, each cellular carrier was allocated 416 channels per simplex band. **If your answer to this question is not 416, you have made an arithmetic slip.**

1.4 The same problem run backwards practice

PROBLEM not a PYQ · AS Sir, Tutorial VIII, Example 2 (alternate form)

In US AMPS, 416 channels are allocated to various cellular operators. The channel between them is 30 kHz with the guard band of 10 kHz. Calculate the spectrum allocation given to each operator.

- **Why this is worth doing:** the papers have only ever asked for N . The same formula rearranged asks for B_t , and **it is the obvious next variant** for a paper to set.
- **Step 1: rearrange the channel-count formula for B_t .** Starting from $N = (B_t - 2B_{\text{guard}})/B_c$, multiply both sides by B_c and add $2B_{\text{guard}}$:

$$B_t = NB_c + 2B_{\text{guard}}$$

- **Step 2: substitute.**

$$B_t = 416 \times (30 \times 10^3) + 2(10 \times 10^3)$$

- **Step 3: evaluate each term.**
 - Channels: $416 \times 30,000 = 12,480,000 \text{ Hz} = 12.48 \text{ MHz}$.
 - Guard bands: $2 \times 10,000 = 20,000 \text{ Hz} = 0.02 \text{ MHz}$.
- **Step 4: add.**

$$B_t = 12,480,000 + 20,000 = 12,500,000 \text{ Hz}$$

- **Final answer: $B_t = 12.5$ MHz for each simplex band.**
- **Check:** this is exactly the input of Problem 1.3, which returned 416. The two problems are inverses of each other, so each one checks the other.

Problem 2 Simultaneously transmitting users in an N-TDMA cluster

PIN 1/23 80 Ch 4

QUESTION AS PRINTED 2080 Chaitra · Q8(b) · [4]

In the N-TDMA system, it uses one way bandwidth of 25 MHz for the forward (or reverse) channel. The system bandwidth is divided into radio channels of 30 kHz. Guard bands with $B_g = 20$ kHz are used. Calculate number of simultaneously transmitting users that can be accommodated in each cluster is.

The trailing “is” is the paper’s own slip. “Cluster” here means the whole system bandwidth, since in a cluster every channel is used exactly once.

- **Step 1: understand what is being counted.** A **cluster** is the group of cells over which the entire allocation is used once, with no repetition. So the number of simultaneously transmitting users per cluster is simply **the number of channels the whole allocation supports**.
 - Frequency reuse means the *next* cluster reuses the same channels, so the answer is per cluster, not per system.
 - **No cluster size N is needed**, and none is given. If the question had asked for users *per cell*, you would then divide by N .
- **Step 2: note which formula applies.** The statement gives no number of time slots per carrier, so take $m = 1$ and the TDMA formula collapses to the FDMA one:

$$N_u = \frac{m(B_s - 2B_g)}{B_c} \xrightarrow{m=1} \frac{B_s - 2B_g}{B_c}$$

- **Step 3: list the data in kHz.**
 - $B_s = 25$ MHz = 25,000 kHz (one-way, so no halving is needed: the statement already says forward or reverse)
 - $B_g = 20$ kHz
 - $B_c = 30$ kHz
- **Step 4: remove both guard bands and divide.**

$$N_u = \frac{25,000 - 2(20)}{30} = \frac{25,000 - 40}{30} = \frac{24,960}{30} = 832$$

- **Final answer: $N_u = 832$ simultaneously transmitting users per cluster.**
- **Sanity check:** the guard bands cost only $40/25,000 = 0.16\%$ of the allocation, so the answer must land just under $25,000/30 = 833.3$. It does.

Problem 3 TDMA frame efficiency in GSM

PIN 2/23 70 Bh 4, 74 Ma 2 · the numbers 156.25, 1250, 322 and 74.24 % are worth memorising outright

3.1 Find the frame efficiency (70 Bh)

QUESTION AS PRINTED 2070 Bhadra · Q8 · [2+6+4]

Define multiple access. What are the merits and demerits of Code Division Multiple Access? If a normal GSM time slot consists of 6 trailing bits, 8.25 guard bits, 26 training bits, and 2 traffic bursts of 58 bits of data, find the frame efficiency.

The first two parts are theory, answered in chapter §7.1 and §7.6. Only the 4-mark calculation is worked here.

- **Step 1: build one time slot, field by field.** Add every field the statement lists. **The 8.25 guard bits are not a typo:** GSM’s guard period really is a quarter bit longer than 8 bits, so the slot length comes out non-integer.

Field	Count	Bits	Overhead or payload?
Trailing (tail) bits	6	6	overhead, ramps the modulator
Guard bits	8.25	8.25	overhead, absorbs propagation delay spread
Training bits	26	26	overhead, trains the equalizer
Traffic bursts	2×58	116	payload , the only useful bits
One time slot		156.25	

- Arithmetic in full: $6 + 8.25 + 26 + 2(58) = 6 + 8.25 + 26 + 116 = 156.25$ bits.
- **Step 2: build the frame.** **A GSM frame holds 8 time slots**, one per user. This number is not in the question, it is standard GSM knowledge you must supply.

$$b_T = 8 \times 156.25 = 1250 \text{ bits per frame}$$

- **Step 3: total the overhead over the whole frame.** Every one of the 8 slots carries its own tail, guard and training bits, so multiply each by 8:

$$b_{OH} = 8(6) + 8(8.25) + 8(26) = 48 + 66 + 208 = 322 \text{ bits}$$

- $8 \times 6 = 48$ (tail)
- $8 \times 8.25 = 66$ (guard)
- $8 \times 26 = 208$ (training)

- **Step 4: apply the frame efficiency formula.** Efficiency is the fraction of the frame that is *not* overhead:

$$\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\% = \left(1 - \frac{322}{1250}\right) \times 100\%$$

- **Step 5: evaluate.**

$$\frac{322}{1250} = 0.2576 \rightarrow 1 - 0.2576 = 0.7424 \rightarrow \eta_f = 74.24\%$$

- **Final answer: $\eta_f = 74.24\%$.**
- **Cross-check the other way.** Payload per frame = $8 \times 116 = 928$ bits, and $928/1250 = 0.7424$. **Same answer, which confirms that “overhead” was defined as everything except the two traffic bursts.**

3.2 Show that TDMA frame efficiency cannot reach 100 % in GSM (74 Ma)

QUESTION AS PRINTED 2074 Magh · Q8 · [6+2]

Explain briefly channel structure of GSM. Show that TDMA frame efficiency cannot reach 100% in GSM

The 6-mark GSM channel structure belongs to Ch 8. Only the 2-mark proof is here.

- **Step 1: write the definition as an inequality, not a number.**

$$\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\%$$

so $\eta_f = 100\%$ requires $b_{OH} = 0$.

- **Step 2: show b_{OH} can never be zero, field by field.** Each field is physically necessary, so none of them can be set to zero:
 - **Guard bits** absorb the difference in propagation delay between a mobile at the cell edge and one beside the base station. Remove them and **bursts from different mobiles overlap**, and both are lost.
 - **Training bits** give the adaptive equalizer a known sequence to estimate the channel from. The channel changes between bursts, so this must be repeated **in every burst**, not sent once.
 - **Tail bits** let the modulator ramp up and down inside the slot. Without them the burst edges splatter energy into adjacent channels.
 - **Preamble / synchronisation bits** let base station and subscriber identify each other and align in time.
- **Step 3: conclude with the actual number.** In GSM, $b_{OH} = 322$ of $b_T = 1250$ bits, i.e. **25.76 % of every frame is overhead**, giving $\eta_f = 74.24\%$. Since $b_{OH} = 322 > 0$, $\eta_f < 100\%$.
- **Final answer: $\eta_f = 74.24\% < 100\%$, and it is bounded away from 100 % because the burst structure that makes TDMA work is itself made of non-payload bits.**
- **The general statement, worth one line: TDMA buys the ability to share a carrier by spending bits on framing.** FDMA transmits continuously and needs almost no framing, which is exactly why FDMA has lower overhead (chapter §7.10).

3.3 GSM slot and frame timing, and simultaneous users practice

PROBLEM not a PYQ · AS Sir, Tutorial VIII, Examples 3 and 4

Consider GSM, a TDMA/FDD system using 25 MHz for the forward link, broken into radio channels of 200 kHz. If 8 speech channels are supported on a single radio channel and no guard band is assumed, find the number of simultaneous users. If each frame consists of 8 time slots, each slot contains 156.25 bits, and data is transmitted at 270.833 kbps, find (a) the time duration of a bit, (b) of a slot, (c) of a frame, and (d) how long a user must wait between two transmissions.

- **Users: step 1, find the bandwidth one user really occupies.** A 200 kHz carrier is shared by 8 users, so **each user's share is $200/8 = 25$ kHz**, even though the user briefly uses the whole 200 kHz.
- **Users: step 2, divide the allocation by that share.** No guard band is assumed, so nothing is subtracted:

$$N = \frac{25 \text{ MHz}}{200 \text{ kHz}/8} = \frac{25 \times 10^6}{25 \times 10^3} = 1000$$

GSM can accommodate 1000 simultaneous users in 25 MHz.

- (a) **Time duration of a bit.** The bit period is the reciprocal of the bit rate:

$$T_b = \frac{1}{270.833 \text{ kbps}} = \frac{1}{270,833} = 3.692 \times 10^{-6} \text{ s} = \mathbf{3.692 \mu s}$$

- (b) **Time duration of a slot.** A slot is 156.25 bits long, so multiply:

$$T_{\text{slot}} = 156.25 \times T_b = 156.25 \times 3.692 \mu s = 576.9 \mu s = \mathbf{0.577 \text{ ms}}$$

- (c) **Time duration of a frame.** A frame is 8 slots:

$$T_f = 8 \times T_{\text{slot}} = 8 \times 0.577 \text{ ms} = \mathbf{4.615 \text{ ms}}$$

- Cross-check through the bit count: $1250 \text{ bits} \times 3.692 \mu s = 4.615 \text{ ms}$. ✓
- So the frame rate is $1/4.615 \text{ ms} = 216.7 \text{ frames per second}$, which also equals $270.833 \text{ kbps}/1250 \text{ bits}$. ✓
- (d) **Wait between transmissions.** A user owns one slot in every frame, so after its burst ends it must wait for the *next* frame to come round: **4.615 ms**, one full frame time.
- **Why this matters:** 4.615 ms of buffering is why TDMA speech has more delay than FDMA speech, and the 0.577 ms burst is why the burst bit rate is 8 times the user's own rate, which is what forces the equalizer (chapter §7.3).

Problem 4 Construct the 8×8 Hadamard code

HOT 3/23 82 Bh **3**, 80 Bh **4**, 79 Ch **3+5**

QUESTION AS PRINTED 2079 Chaitra · Q7 **3+5** · also 2082 Bhadra Q6(a) **2+3**, 2080 Bhadra Q6(b) **4**

Write the steps to satisfy condition for Hadamard code. Construct 8 by 8 Hadamard code that satisfies above conditions.

2082 Bhadra prints “Define turbo coding. Construct 8 by 8 Hadamard code.” and 2080 Bhadra prints “Construct Hadamard matrix for H_8 .” Same construction, three different mark splits.

Step 1 State the conditions the matrix must satisfy **3**

- It is a **square** matrix of order N , every entry $+1$ or -1 .
- **Every two distinct rows are orthogonal:** multiply them element by element, add up the products, and the result is 0.
- Equivalently $H_N H_N^T = N I_N$, where I_N is the $N \times N$ identity.
- The order must be $N = 1$, $N = 2$, or a **multiple of 4**. There is no 3×3 or 6×6 Hadamard matrix.
- The N rows are the N **codewords**, each N bits long, so the code carries $k = \log_2 N$ information bits. For $N = 8$: **8 codewords, 8 bits each, $k = 3$** .
- Minimum distance is $d_{\min} = N/2 = 4$, so it detects 3 errors and corrects $\lfloor (4 - 1)/2 \rfloor = 1$ error.

Step 2 Start from H_1 and H_2

- The seed is the trivial 1×1 matrix $H_1 = [1]$.
- Applying the recursion once gives the 2×2 matrix everybody starts from:

$$H_2 = \begin{bmatrix} H_1 & H_1 \\ H_1 & -H_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- **Check it:** row 1 times row 2 = $(1)(1) + (1)(-1) = 1 - 1 = 0$. ✓ Orthogonal.

Step 3 Double it to H_4 using the Sylvester recursion

- The rule is $H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix}$: **copy the block three times and negate the bottom right one.**
- $-H_2$ means flip the sign of every entry: $-H_2 = \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$.
- Assembling the four blocks:

$$H_4 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

- **Check a pair:** row 2 times row 3 = $(1)(1) + (-1)(1) + (1)(-1) + (-1)(-1) = 1 - 1 - 1 + 1 = 0$. ✓

Step 4 Double once more to H_8 **5**

- Apply the same rule to H_4 : $H_8 = \begin{bmatrix} H_4 & H_4 \\ H_4 & -H_4 \end{bmatrix}$.
- The top-left, top-right and bottom-left 4×4 blocks are all H_4 copied unchanged. The bottom-right block is H_4 with every sign flipped.

$$H_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix}$$

Step 5 Verify the orthogonality condition

- Pick any two distinct rows, multiply element by element, add. **Show at least one such check in the exam**, it is worth a mark.
- Row 2 against row 3:

$$\begin{aligned} (1)(1) + (-1)(1) + (1)(-1) + (-1)(-1) + (1)(1) + (-1)(1) + (1)(-1) + (-1)(-1) \\ = 1 - 1 - 1 + 1 + 1 - 1 - 1 + 1 = 0 \quad \checkmark \end{aligned}$$

- Row 1 against row 5:

$$1 + 1 + 1 + 1 + 1 - 1 - 1 - 1 - 1 = 0 \quad \checkmark$$

- Any row against itself: $1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 8 = N$. $\checkmark$ This is the second property CDMA decoding uses.

Step 6 Convert to binary Walsh codes

- For transmission the ± 1 entries are mapped to bits: **+1 $\rightarrow$ 0 and $-1 \rightarrow$ 1**.
- That turns the eight rows of H_8 into the eight **Walsh codes** $W_0 \dots W_7$:

Code	Binary	Ones	Code	Binary	Ones
W_0	0 0 0 0 0 0 0 0	0	W_4	0 0 0 0 1 1 1 1	4
W_1	0 1 0 1 0 1 0 1	4	W_5	0 1 0 1 1 0 1 0	4
W_2	0 0 1 1 0 0 1 1	4	W_6	0 0 1 1 1 1 0 0	4
W_3	0 1 1 0 0 1 1 0	4	W_7	0 1 1 0 1 0 0 1	4

- **Check:** every codeword except the all-zero one has exactly $N/2 = 4$ **ones**, which is the binary statement of $d_{\min} = N/2$.
- **Scaling up:** one more application of the recursion gives H_{64} , whose 64 rows are what IS-95 actually uses. W_0 , the all-zero row, is its pilot channel. See chapter §7.7 for where these codes sit in the CDMA transmitter.

Problem 5 CDMA encoding and decoding, worked with numbers

HOT 3/23 82 Bh **7**, 79 Ch **8**, 77 Ch **6** · the paper always says “with relevant example”, so an answer with no arithmetic loses most of the marks

QUESTION AS PRINTED 2082 Bhadra · Q7(a) **7** · 2079 Chaitra Q8(a) **8** · 2077 Chaitra Q10(a) **2+6**

Explain the implementation of CDMA with its encoding part and decoding part with relevant example.

2079 Chaitra prints “Illustrate Encoding and decoding mechanism of CDMA system in brief.” 2077 Chaitra prints “What are the major disadvantages of spread spectrum system? Explain the implementation of CDMA system with appropriate example.” The 2-mark disadvantages rider is answered in chapter §7.4 and §7.6.

5.1 Four stations, all transmitting

- **Step 1: write down the rules the arithmetic uses.** State these before calculating, they are worth marks on their own.
 1. Codes are **orthogonal**: $C_i \cdot C_j = 0$ for $i \neq j$.
 2. A code times itself gives $C_i \cdot C_i = N$, the number of stations. Here $N = 4$.
 3. **Data mapping**: bit 1 $\rightarrow$ +1, bit 0 $\rightarrow$ -1, silent $\rightarrow$ 0.
 4. Multiplying a sequence by a number scales every element of it.
- **Step 2: set up the given data.** Four stations, data $d_1 = 1$, $d_2 = 0$, $d_3 = 1$, $d_4 = 1$, and the four orthogonal chip sequences:

Station	Chip sequence C_i	Data bit d_i	Mapped value
ST1	$C_1 = (+1, +1, +1, +1)$	1	+1
ST2	$C_2 = (+1, -1, +1, -1)$	0	-1
ST3	$C_3 = (+1, +1, -1, -1)$	1	+1
ST4	$C_4 = (+1, -1, -1, +1)$	1	+1

- **Step 3: confirm the codes really are orthogonal** before using them. Two spot checks are enough in the exam:
 - $C_1 \cdot C_2 = (1)(1) + (1)(-1) + (1)(1) + (1)(-1) = 1 - 1 + 1 - 1 = 0 \checkmark$
 - $C_3 \cdot C_4 = (1)(1) + (1)(-1) + (-1)(-1) + (-1)(1) = 1 - 1 + 1 - 1 = 0 \checkmark$
 - $C_1 \cdot C_1 = 1 + 1 + 1 + 1 = 4 = N \checkmark$
- **Step 4: encode. Each station multiplies its own code by its own data value.**
 - $d_1 C_1 = (+1)(+1, +1, +1, +1) = (+1, +1, +1, +1)$
 - $d_2 C_2 = (-1)(+1, -1, +1, -1) = (-1, +1, -1, +1)$
 - $d_3 C_3 = (+1)(+1, +1, -1, -1) = (+1, +1, -1, -1)$
 - $d_4 C_4 = (+1)(+1, -1, -1, +1) = (+1, -1, -1, +1)$
- **Step 5: all four transmit at once, and the channel adds them.** Sum the four sequences **element by element**, i.e. column by column:

Contribution	chip 1	chip 2	chip 3	chip 4
$d_1 C_1$	+1	+1	+1	+1
$d_2 C_2$	-1	+1	-1	+1
$d_3 C_3$	+1	+1	-1	-1
$d_4 C_4$	+1	-1	-1	+1
Common channel S	+2	+2	-2	+2

- Column by column: $1 - 1 + 1 + 1 = +2$, $1 + 1 + 1 - 1 = +2$, $1 - 1 - 1 - 1 = -2$, $1 + 1 - 1 + 1 = +2$.
- So the composite signal on the common channel is $S = (+2, +2, -2, +2)$.
- **Step 6: decode. Suppose ST3 wants ST2's data.** Multiply S by ST2's code element by element, add all four products, then divide by $N = 4$:

$$S \times C_2 = (+2)(+1), (+2)(-1), (-2)(+1), (+2)(-1) = (+2, -2, -2, -2)$$

$$\text{sum} = 2 - 2 - 2 - 2 = -4 \quad \rightarrow \quad \frac{-4}{4} = -1$$

- Applying the mapping backwards, -1 means **bit 0**, which is exactly d_2 . $\checkmark$
- **Step 7: show the same receiver recovers every station.** This is the proof the scheme works, and it is where the last marks are:

Decode with	$S \times C_i$, element by element	Sum	$\div 4$	Recovered bit
C_1	$(+2, +2, -2, +2)$	+4	+1	1 = $d_1 \checkmark$
C_2	$(+2, -2, -2, -2)$	-4	-1	0 = $d_2 \checkmark$
C_3	$(+2, +2, +2, -2)$	+4	+1	1 = $d_3 \checkmark$
C_4	$(+2, -2, +2, +2)$	+4	+1	1 = $d_4 \checkmark$

- **Final answer: all four data bits (1, 0, 1, 1) are recovered from one shared 4-chip sequence**, with every station using the whole bandwidth for the whole time.
- **Why it works, in one line:** in $S \times C_k$ every cross term $d_i(C_i \cdot C_k)$ with $i \neq k$ is zero by orthogonality, so only $d_k(C_k \cdot C_k) = d_k N$ survives. Dividing by N returns d_k .

5.2 Three stations, one of them silent

PROBLEM not a PYQ · SST Ch 7, Figs 12.34 and 12.35 · the textbook version of the same question

Stations A, B and C are assigned codes $C_A = (+1, -1, -1, +1)$, $C_B = (+1, +1, +1, +1)$ and $C_C = (+1, -1, +1, -1)$. Station A wants to transmit a 0 bit, station B wants to remain silent, and station C wants to transmit a 1. Find the multiplexed CDMA signal, and show that a demultiplexer recovers all three states.

- **Why work this one too:** it is the version drawn in the lecture figures, and it adds the **silent station** case, which the four-station example does not cover. A paper asking “with relevant example” accepts either.
- **Step 1: map the three states.** Bit 0 $\rightarrow -1$, **silent** $\rightarrow 0$, bit 1 $\rightarrow +1$.

- A: bit 0 $\rightarrow -1$
- B: silent $\rightarrow 0$
- C: bit 1 $\rightarrow +1$
- **Step 2: encode each station.**
 - $-1 \cdot C_A = (-1)(+1, -1, -1, +1) = (-1, +1, +1, -1)$
 - $0 \cdot C_B = (0)(+1, +1, +1, +1) = (0, 0, 0, 0)$ **a silent station contributes nothing at all**
 - $+1 \cdot C_C = (+1)(+1, -1, +1, -1) = (+1, -1, +1, -1)$
- **Step 3: add element by element to get the multiplexed signal.**
 - chip 1: $-1 + 0 + 1 = 0$
 - chip 2: $+1 + 0 - 1 = 0$
 - chip 3: $+1 + 0 + 1 = +2$
 - chip 4: $-1 + 0 - 1 = -2$

$$S = (0, 0, +2, -2)$$

- **Step 4: demultiplex each branch.** Multiply, add all bits, divide by $N = 4$:

Branch	$S \times C_i$, element by element	Add all bits	$\div 4$	Result
A, C_A	(0, 0, -2, -2)	-4	-1	bit 0 ✓
B, C_B	(0, 0, +2, -2)	0	0	silent ✓
C, C_C	(0, 0, +2, +2)	+4	+1	bit 1 ✓

- **Final answer:** $S = (0, 0, +2, -2)$, and the decoder returns **bit 0, silent, bit 1**, matching what the three stations sent.
- **Note the three-way result.** The decoder does not just distinguish 1 from 0. It also reports *silence* as an exact zero, which is what lets CDMA gain capacity from **voice activity**: a silent user costs the system nothing.

Problem 6 Capacity of an IS-95 CDMA cell practice

No paper 2070–2082 has set this, but it is the one calculation the CDMA theory implies and it is a standard tutorial problem. Included so the pattern is covered.

PROBLEM not a PYQ · AS Sir, Tutorial VIII, Example 8

In an IS-95 CDMA system, if $W = 1.25$ MHz, $R = 9600$ bps and $N = 14$ users, (a) calculate E_b/N_0 ; (b) when there is no voice activity, calculate it for omnidirectional antennas; (c) if voice activity = $3/8$ and 3-sector antennas are used, calculate the total number of users per cell.

- **Step 1: find the processing gain.** It is the ratio of spread bandwidth to data rate:

$$\frac{W}{R} = \frac{1.25 \times 10^6}{9600} = 130.21$$

- **(a) Step 2: apply the single-cell CDMA capacity relation.** Each of the other $N - 1$ users contributes an equal share of interference, so

$$\frac{E_b}{N_0} = \frac{W/R}{N-1} = \frac{130.21}{14-1} = \frac{130.21}{13} = 10.02$$

- In decibels, $10 \log_{10}(10.02) = 10.01$ dB. **Final answer:** $E_b/N_0 \approx 10$, i.e. **about 10 dB**.
 - The linear value 10.02 and the dB value 10.01 coincide here only because $10 \log_{10} 10 = 10$. Do not assume that in other problems.
- **(b) With no voice activity and omnidirectional antennas, nothing changes.** Voice activity factor $\alpha = 1$ means every user talks all the time, which is the assumption already built into part (a). An omnidirectional antenna gives no sectorisation gain. **So E_b/N_0 stays at 10.02, about 10 dB.**
- **(c) Step 3: rearrange the relation for N and add the voice activity factor.** Solving $E_b/N_0 = (W/R)/(N-1)$ for N gives $N = 1 + (W/R)/(E_b/N_0)$. A user who is silent a fraction of the time contributes interference only while talking, so the term is divided by α :

$$N = 1 + \frac{1}{\alpha} \left[\frac{W/R}{E_b/N_0} \right]$$

- **Step 4: substitute $\alpha = 3/8$, so $1/\alpha = 8/3$:**

$$N = 1 + \frac{8}{3} \left[\frac{130.21}{10} \right] = 1 + \frac{8}{3}(13.021) = 1 + 34.72 = 35.72$$

i.e. about **35.7 users per sector**.

- **Step 5: apply the sectorisation gain.** A 3-sector antenna splits the cell into three, and each sector only hears the interference of its own users, so the cell carries three times the per-sector figure:

$$N_{\text{cell}} = 3 \times 35.72 = 107.2$$

- **Final answer: about 107 users per cell.**
- **Reading the result:** the raw processing gain alone would allow 14 users. Voice activity multiplies that by $8/3$ and sectoring by 3, taking it to roughly 107. **This is exactly the “soft capacity” argument of chapter §7.6 in numbers:** CDMA capacity is set by the interference budget, not by a fixed count of channels.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver

Short	Means	Short	Means
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access

Short	Means	Short	Means
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK, DBPSK	differential (binary) PSK
CPFSK	continuous phase FSK	MFSK, MPSK	M-ary FSK, M-ary PSK
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB, VSB	double, single, vestigial sideband
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	<i>BT</i>	GMSK filter bandwidth $\times$ symbol time

Short	Means	Short	Means
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
SSMA	spread spectrum multiple access	SDMA	space division multiple access
FHMA	frequency hopped multiple access	DSMA	direct sequence multiple access (= CDMA)
FCDMA	hybrid FDMA/CDMA	DS/FHMA	hybrid direct sequence / frequency hopped
TCDMA	time division CDMA	TDFH	time division frequency hopping
IM	intermodulation (product)	MU-MIMO	multi-user MIMO
WLAN	wireless local area network	WMAN	wireless metropolitan area network
WPAN	wireless personal area network	AP	access point
BSS, IBSS	basic service set, independent (ad hoc) BSS	ESS	extended service set
CSMA/CA	carrier sense multiple access with collision avoidance	CSMA/CD	... with collision detection
DCF	distributed coordination function	PCF	point coordination function
RTS/CTS	request to send / clear to send	CCK	complementary code keying
WEP, WPA	wired equivalent privacy, WiFi protected access	U-NII	unlicensed national information infrastructure band
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.

Short	Means	Short	Means
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
BSS	base station subsystem	NSS	network and switching subsystem
OSS	operation support subsystem	OMC	operation and maintenance centre
HLR	home location register	VLR	visitor location register
AuC	authentication centre	EIR	equipment identity register
CEIR	central EIR	ME	mobile equipment
SIM	subscriber identity module	IMSI	international mobile subscriber identity
Short	Means	Short	Means
IMEI	international mobile equipment identity	TMSI	temporary mobile subscriber identity
Um, Abis, A	the three GSM interfaces	SS7	signalling system no. 7
ARFCN	absolute radio frequency channel number	TN, TS	time slot number, time slot
TCH	traffic channel	TCH/FS, TCH/HS	full rate, half rate speech TCH
CCH	control channel	CCCH	common control channel
DCCH	dedicated control channel	SDCCH	stand-alone dedicated control channel
SACCH	slow associated control channel	FACCH	fast associated control channel
FCCH	frequency correction channel	SCH	synchronisation channel
PCH	paging channel	RACH	random access channel
AGCH	access grant channel	TB, GP, SF	tail bits, guard period, stealing flag
VAD	voice activity detector	DTX	discontinuous transmission
DCS 1800	digital cellular system, the 1800 MHz GSM	CRC	cyclic redundancy check
ESN	electronic serial number	PCB	power control bit
Mcps, Ksps	mega-chips, kilo-symbols per second	PRAT	paging channel rate field
EVRC	enhanced variable rate coder	PDSN	packet data serving node
PCF (CDMA)	packet control function	RNC	radio network controller
Short	Means	Short	Means
EPS, EPC	evolved packet system, core	eNodeB	LTE base station
MME	mobility management entity	S-GW, P-GW	serving, packet data network gateway
HSS	home subscriber server	PCRF	policy and charging rules function
IMS, VoLTE	IP multimedia subsystem, voice over LTE	CoMP	coordinated multipoint
ITU-R	ITU radiocommunication sector	WRC	world radiocommunication conference
NTA	Nepal Telecommunications Authority	UGS, rtPS, nrtPS	WiMAX QoS service classes

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

Three clashes to watch: **FCC** is the *forward control channel* throughout these notes; the US regulator is always written out as the Federal Communications Commission, and **FCCH** is GSM's frequency correction channel, a different thing again. **BCH** is the Bose–Chaudhuri–Hocquenghem block code in **Ch 6** and GSM's *broadcast channel* in **Ch 8**. **SCH** is GSM's synchronisation channel, never a sub-channel.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 23 papers, 2070–2082 BS: **TOP /23** = topic asked in 6 or more papers · **HOT /23** = 3 to 5 papers · **PIN /23** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.